
AZ-204 Class Notes 202X

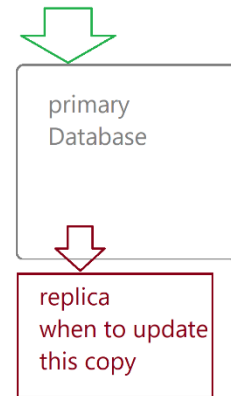
Contents

Links for AZ-204	2
Code – Dotnet – Az CLI – PowerShell – C# - Syntax – Script - Snippets	3
Syntax: What Gets Typed & Special Characters.....	3
Code Talk: Coding, Scripting – what gets used where from exam point of view.....	4
AZ-204 Labs Table of Contents & Tim's Notes.....	12
Cosmos DB	14
Change Feed.....	14
How do relationships get created?	15
PowerShell on Local Windows	30
Data Movement - Storage.....	33
Commands Code Speak Cross Reference	42
 When to Use What?	73
Service Bus Message Unit & Billing.....	74
 What Exactly Is Event Hubs?	80
Azure Storage Queues vs. Service Bus Queues.....	81
Containers Including Docker	82
Azure Container Registry Tasks (ACR Tasks)	82
 Azure Container Instances (ACI)	82
 Azure Container Apps (ACA).....	83
Deployment Slots Swapping	83
What is Microsoft Graph?	84
Authentication MSAL.NET	86
Reference	86

Links for AZ-204 – 05/06/2024 – 12/03/2025

1. [Course AZ-204T00-A: Developing Solutions for Microsoft Azure - Training | Microsoft Learn](#)
 - a. Official Book
2. [GitHub - MicrosoftLearning/AZ-204-DevelopingSolutionsforMicrosoftAzure: AZ-204: Developing solutions for Microsoft Azure](#)
 - a. Lab files & Instructions
3. [Microsoft Certified: Azure Developer Associate - Certifications | Microsoft Learn](#)
 - a. AZ-204 Exam Link
4. [Study guide for Exam AZ-204: Developing Solutions for Microsoft Azure | Microsoft Learn](#)

Consistency Level	Description
Strong saved	When writes are performed on your primary database, the operation is replicated to the replica instances. The write operation is committed (and visible) on the primary only after it has been committed and confirmed by all replicas. ↓ record or row
Bounded Staleness	Like the Strong level with the difference that you can configure how stale documents can be within replicas. Staleness is the quantity of time (or the version count) a replica document can be behind the primary document. ↑ ↑
Session	This level guarantees that all read and write operations are consistent within a user session. Within the user session, all reads and writes are monotonic and guaranteed to be consistent across primary and replica instances.
Consistent Prefix	This level has loose consistency but guarantees that when updates show up in replicas they will show up in the correct order (that is, as prefixes of other updates) without any gaps.
Eventual	This level has the loosest consistency and essentially commits any write operation against the primary immediately. Replica transactions are asynchronously handled and will eventually (over time) be consistent with the primary. This tier has the best performance, because the primary database does not need to wait for replicas to commit to finalize its transactions.



Code – Dotnet – Az CLI – PowerShell – C# - Syntax – Script - Snippets

5. [dotnet command - .NET CLI | Microsoft Learn](#) - SQL
6. [Azure Command-Line Interface \(CLI\) documentation | Microsoft Learn](#)
7. [PowerShell Documentation - PowerShell | Microsoft Learn](#)

8. Login dup

them in the labs as needed.

Instructions

Before you start

Sign in to the lab environment

Sign in to your Windows 11 virtual machine (VM) by using the following credentials:

- Username: [T] Admin
- Password: [T] Pa55w.rd

Note: Your instructor will provide instructions to connect to the virtual lab environment.

Review the installed applications

Find the taskbar on your Windows 11 desktop. The taskbar contains the icons for the applications that you'll use in this lab, including:

- Microsoft Edge
- File Explorer
- Terminal

9.

10. endList

Syntax: What Gets Typed & Special Characters

In C#, the **<BlobContainerClient>** part in `private async Task<BlobContainerClient>` is a **generic type parameter**.

Breaking it down:

- `Task<T>` is a **generic class** in the `.NET System.Threading.Tasks` namespace that represents an **asynchronous operation** that returns a result of type `T`.
- In your example:
- `private async Task<BlobContainerClient> SomeMethodName()`

This means:

- The method is private (accessible only in this class).
- It's async (can use `await` and returns asynchronously).
- It returns a `Task<BlobContainerClient>`, meaning:
 - The method runs asynchronously.
 - When awaited, it will eventually return a `BlobContainerClient` object.

So what are `< >` in this context?

They're **type parameters** used in **generics**, telling the compiler what type to substitute for `T`.

A few more examples for clarity:

Syntax	Meaning
List<string>	A list of strings
Dictionary<int, string>	A dictionary with int keys and string values
Task<int>	An async method returning an int
IEnumerable<MyClass>	An enumerable collection of MyClass instances

Let me know if you want a visual or deeper dive into generics!

```
private async Task<BlobContainerClient> GetCloudBlobContainer(string containerName)
```

private async Task<BlobContainerClient>
- what are the greater than & less than symbols in c# syntax

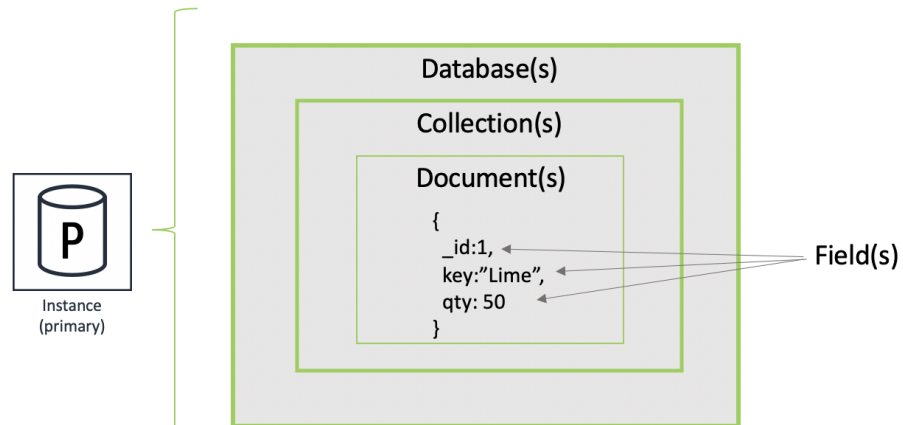
The ellipsis symbol (...) is a punctuation mark consisting of three periods (or dots) used to indicate an omission of words, a pause, or a trailing off of thought. It can also indicate an unfinished sentence or a trailing off into silence, according to Wikipedia. In some contexts, particularly in quoted material, it signifies that some text has been omitted. 



Code Talk: Coding, Scripting – what gets used where from exam point of view...

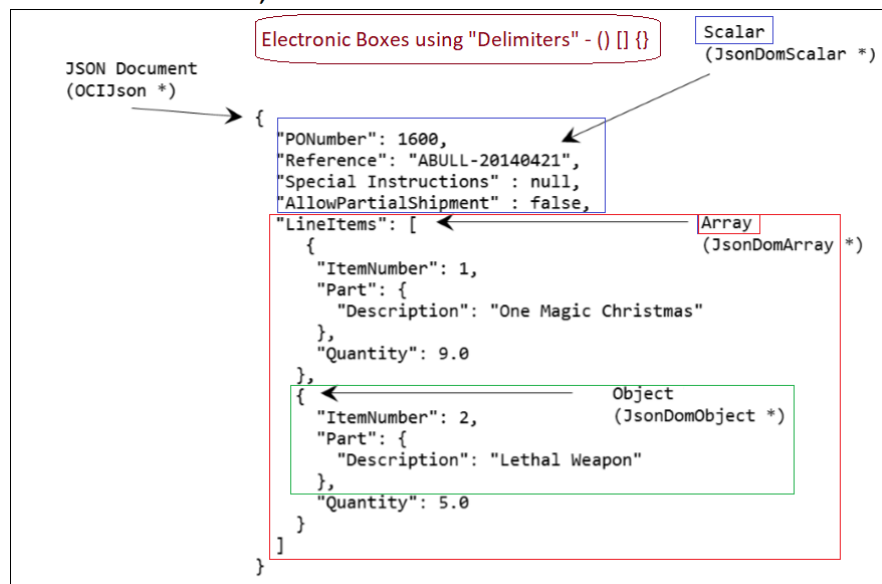
JavaScript, SQL, C#, PowerShell, Bash, cmd shell, (Azure CLI as a family name) & dotnet for starters...

1. Creating Multi-model database – (creating data relations and working with that data)
 - a. Document JSON & CosmosDB
 - b. [Documents - Azure Cosmos DB REST API | Microsoft Learn](#)



id - Required. It is a user settable property and is the unique name that identifies the attachment, that is, no two attachments share the same ID

c.



d.

```
select * from c IN
```

Box A

```
where
```

Box B

```
= "9.99"
```

```
{
  "id": "1",
  "Name": "UserA",
  "orders": [
    {
      "course": "Big Data",
      "price": "9.99"
    }
  ],
  "ratings": { "3": "100", "4": "200", "5": "300" }
}
```

e.

2. [CloudBlockBlob Class \(Microsoft.Azure.Storage.Blob\) - Azure for .NET Developers | Microsoft Learn](#)

```
CloudBlockBlob montanacloudBlockBlob =
cloudBlobContainer.GetBlockBlobReference(imgName);
await montanacloudBlockBlob.UploadFromFileAsync(imgFile);
```

3.

```
C#
// Create a file in your local MyDocuments folder to upload to a blob.
string localPath = Environment.GetFolderPath(Environment.SpecialFolder.MyDocuments);
string localFileName = "QuickStart_" + Guid.NewGuid().ToString() + ".txt";
string sourceFile = Path.Combine(localPath, localFileName);
// Write text to the file.
File.WriteAllText(sourceFile, "Hello, World!");

Console.WriteLine("Temp file = {0}", sourceFile);
Console.WriteLine("Uploading to Blob storage as blob '{0}'", localFileName);

// Get a reference to the blob address, then upload the file to the blob.
// Use the value of localFileName for the blob name.
CloudBlockBlob cloudBlockBlob = cloudBlobContainer.GetBlockBlobReference(localFileName);
await cloudBlockBlob.UploadFromFileAsync(sourceFile);
```

4.

Scripting languages for Workflow, Tasks
vs. Object Oriented Programming an app...

Command	Azure CLI	Azure PowerShell
Create Resource Group	<pre>az group create --name <ResourceGroupName> -- location eastus</pre>	<pre>New-AzResourceGroup -Name <ResourceGroupName> - Location eastus</pre>

-Use PowerShell to install Azure CLI to make the AZ
(Bash) commands run

Azure PowerShell

```
Get-Command -Module Az.*
```

- cmd.exe on Windows
- Bash on Linux or macOS

Azure CLI

```
az webapp --help
```

-CLI commands start with az

.NET CLI

```
dotnet build-server shutdown -h|--help
dotnet build-server -h|--help
```

-dotnet new

-Navigate folders with cd, dir, md, ..

5.

JavaScript

Fill in the blank Q - AZ-204

Copy

```
function validateToDoItemTimestamp() {
  var context = getContext();
  var request = context.getRequest();

  // item to be created in the current operation
  var itemToCreate = request.getBody();

  // validate properties
  if (!("timestamp" in itemToCreate)) {
    var ts = new Date();
    itemToCreate["timestamp"] = ts.getTime();
  }

  // update the item that will be created
  request.setBody(itemToCreate);
}
```

```
function OrderTip() {
  var r = getContext().getRequest();

  var is=r.getBody();

  if(!("Ordertip" in i)) {
    i[tip]=0;
  }

  r.setBody(i);
}
```

6.

7. PARAMETERS:

- Specify which values you can input when deploying the resources.
- Reduce the number of parameters by using objects with multiple properties.

8. VARIABLES: (JSON vs. JavaScript)

- Construct values that can be used throughout your template.
- Define variables is not required but can simplify your template by reducing complex expressions.

```

"parameters": {
  "storageAccountType": {
    "type": "string",
    "defaultValue": "Standard_LRS",
    "allowedValues": [
      "Standard_LRS",
      "Standard_GRS",
      "Standard_ZRS",
      "Premium_LRS"
    ],
    "metadata": {
      "description": "Storage Account type"
    }
  },
  //Declare variable or a parameter in JSON:
  //This one is named location - inside the "parameters": curly
  //brace
  "location": {
    "type": "string",
    "defaultValue": "",
    "metadata": {
      "description": "Location for all resources."
    }
  }
},
"variables": {
  "storageAccountName": "[concat('storage', uniquestring(
    resourceGroup().id))]"
},

```

inside curly brace

The curly brace delimiters create a "box" for this code block -called parameters:

JSON: JavaScript Object Notation
NOT JavaScript
var, let, const variable declarations

name

Create reusable values for deploying the resources here.

variables are used throughout your template but not intended to work with "outside" values like Azure storageAccountType or location

-
-
-
- JavaScript

```

--Psuedo code - means its all talk no execution...

//declare JavaScript variables p1, p2 and give them a value
//1 is the "argument" for p1
var p1 = 1

//2 is the "argument" for p2
var p2 = 2

//pass variable to function parameters p1, p2
function(p1, p2){
  return p1 + p2
}

```

-
-
-
-
-

9. FUNCTIONS:

```
JSON
{
  "functions": [
    {
      "namespace": "<namespace-for-functions>",
      "members": {
        "<function-name>": {
          "parameters": [
            {
              "name": "<parameter-name>",
              "type": "<type-of-parameter-value>"
            }
          ],
          "output": {
            "type": "<type-of-output-value>",
            "value": "<function-return-value>"
          }
        }
      }
    }
  ]
}
```

- The function can't access variables.
- The function can only use parameters that are defined in the function. When you use the [parameters function](#) within a user-defined function, you're restricted to the parameters for that function.
- The function can't call other user-defined functions.
- The function can't use the [reference function](#).
- Parameters for the function can't have default values.

a.

Birds Eye

ARM Templates

Subscription named Subscription1

Resource Group name RG1 in in North Europe Azure region.



Create a Storage account in RG1 with same location as RG1

Subscription1

 RG1 North Europe Azure
 Create a Storage

Select the ideal location value:

 North Europe
 "[resourceGroup().location]"
 "[resourceGroup().region]"
 "[RG1().location]"

```
{
  "location": {
    "type": "string",
    "defaultValue": ,
    "metadata": {
      "description": "Location for all resources."
    }
  }
}
```



Question - Answer

```
{
  "location": {
    "type": "string",
    "defaultValue": "[resourceGroup().location]",
    "metadata": {
      "description": "Location for all resources."
    }
  }
}
```



10.

Azure CLI

```

--Note order of syntax - cart before horse - group, appservice plan, app, deployment
has Git info out of order - incorrect

az group create --location westeurope --name myResourceGroup
az webapp create --name $webappname --resource-group myResourceGroup --plan
$webappname
az appservice plan create --name $webappname --resource-group myResourceGroup --sku
FREE

az webapp deployment source config --name $webappname --resource-group
myResourceGroup --repo-url $gitrepo --branch master --manual-integration

----- correct
az group create --location westeurope --name myResourceGroup
az appservice plan create --name $webappname --resource-group myResourceGroup --sku
FREE
deployment source vs. create source
az webapp create --name $webappname --resource-group myResourceGroup --plan
$webappname
az webapp deployment source config --name $webappname --resource-group
myResourceGroup --repo-url $gitrepo --branch master --manual-integration

----- (az webapp create source myResourceGroup --repo-url) (--
repo-url $gitrepo --branch master --manual-integration)

az group create --location westeurope --name myResourceGroup

az appservice plan create --name $webappname --resource-group myResourceGroup --sku
FREE
az webapp create --name $webappname --resource-group myResourceGroup --plan
$webappname
incorrect
az webapp create source config --name $webappname --resource-group myResourceGroup
--repo-url $gitrepo --branch master --manual-integration

----- (git clone $gitrepo) incorrect

az group create --location westeurope --name myResourceGroup

az appservice plan create --name $webappname --resource-group myResourceGroup --sku
FREE
az webapp create --name $webappname --resource-group myResourceGroup --plan
$webappname
az webapp deployment source config --name $webappname --resource-group
myResourceGroup git clone $gitrepo

```

11.

a.

Dotnet

1. [.NET Tutorial | Hello World in 5 minutes \(microsoft.com\)](#)
2. Example: dotnet new console -o MyApp -f net7.0

Tools for execution of commands through a terminal using interactive command-line prompts or a script vs. writing an app in C# or JavaScript. (Exams may or may not mention these)

cmd.exe or PowerShell on Windows - Bash on Linux or macOS

- Azure Cloud Shell (runs AZ CLI & PowerShell)
- Terminal (runs Command Prompt, AZ CLI & PowerShell in separate tabs)
- Visual Studio Code (has built in Terminal)
- Command Prompt
- PowerShell
- WSL

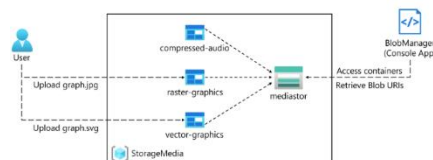
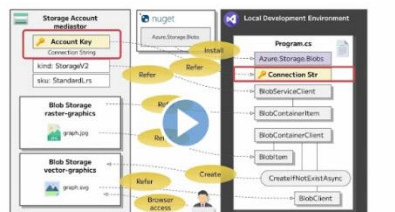
- Windows Subsystem for Linux (WSL) is a feature of Windows that allows developers to run a Linux environment
- [Windows Terminal SSH | Microsoft Learn](#)
 - a connectivity tool for remote sign-in (see also RDP for Windows)
- Azure Cloud Shell Connector



RHH (Rabbit Hole Handling)

AZ-204 Labs Table of Contents & Tim's Notes

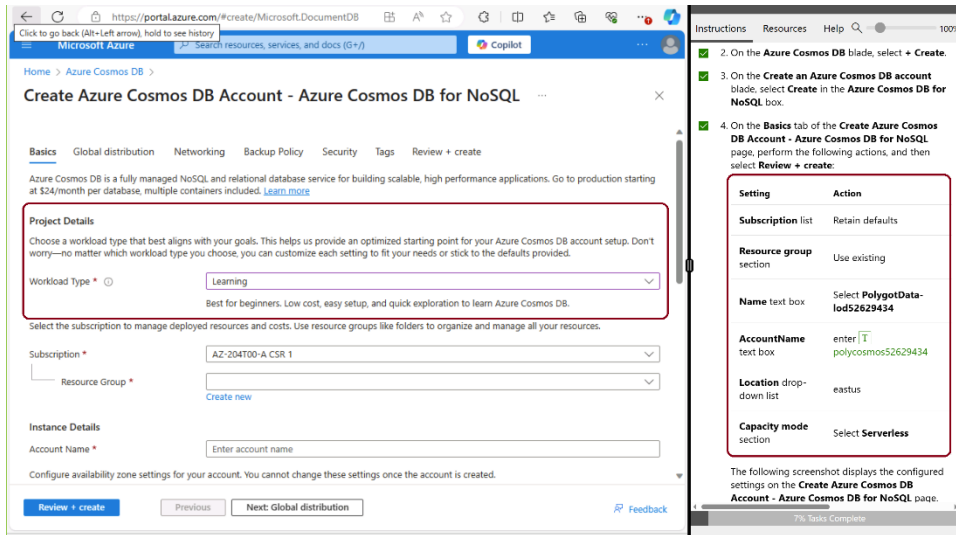
1. Module 01: Creating Azure App Service Web Apps
 - a. Create a web application on Azure by using the PaaS model. After the web application is created, you will learn how to upload existing web application files by using the Apache Kudu zip deployment option.
 - b. Imgapi
 - c. imgweb
2. Module 02: Implement Azure Functions vs. Durable w/Orchestration
 - a. Create a simple Azure function that echoes text that is entered and sent to the function by using HTTP POST commands.
 - b. Connect to a storage account that you create and return the contents of a file that is stored in the Azure storage account
 - c. Execute on a fixed schedule. – CRON job is a timer or an Agent in MS Windows
 - d. The function will write a message to a log each time the schedule is triggered.
 - e. Create a new local Azure Functions project in the current directory using the **dotnet-isolated** runtime:
 - f. [Develop Azure Functions locally using Core Tools | Microsoft Learn](#)
 - g. [Create your first function in the Azure portal | Microsoft Learn](#)
 - h. [Create a C# function using Visual Studio Code - Azure Functions | Microsoft Learn](#)
 - i. [Azure Functions Core Tools reference | Microsoft Learn](#)
 - j. [Create a function that integrates with Azure Logic Apps | Microsoft Learn](#)
 - k. [Test web APIs with the HttpRepl | Microsoft Learn](#)
 - l. funclogic – Function App
3. Module 03: Develop solutions that use blob storage
 - a. Use the Azure Storage SDK to access Azure Storage containers within a C# application – C.R.U.D. Ops.
 - b. Use the Azure Storage SDK to access Azure Storage containers within a C# application.
 - c. Access metadata and expose URI information to gain access to the contents of the containers in the storage account



- d.
- e. BlobManager - console
- f. [Explore Azure Storage security features - Training | Microsoft Learn](#)
- g. [Manage the Azure Blob storage lifecycle - Training | Microsoft Learn](#) – access tiers

- h. [Work with Azure Blob storage - Training | Microsoft Learn](#) – methods

Cosmos DB



Project Details

Choose a workload type that best aligns with your goals. This helps us provide an optimized starting point for your Azure Cosmos DB account setup. Don't worry—no matter which workload type you choose, you can customize each setting to fit your needs or stick to the defaults provided.

Workload Type: Learning

Best for beginners. Low cost, easy setup, and quick exploration to learn Azure Cosmos DB.

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription: AZ-204T00-A CSR 1

Resource Group: Create new

Instance Details

Account Name: Enter account name

Configure availability zone settings for your account. You cannot change these settings once the account is created.

Review + create Previous Next: Global distribution Feedback

Instructions

2. On the **Azure Cosmos DB** blade, select **Create**.
3. On the **Create an Azure Cosmos DB account** blade, select **Create** in the **Azure Cosmos DB for NoSQL** box.
4. On the **Basic** tab of the **Create Azure Cosmos DB Account - Azure Cosmos DB for NoSQL** page, perform the following actions, and then select **Review + create**:

Setting	Action
Subscription list	Retain defaults
Resource group section	Use existing
Name text box	Select PolygotData-Iod52629434
AccountName text box	enter polycosmos52629434
Location drop-down list	eastus
Capacity mode section	Select Serverless

The following screenshot displays the configured settings on the **Create Azure Cosmos DB Account - Azure Cosmos DB for NoSQL** page.

7% Tasks Complete

4. Module 04: Develop solutions that use Cosmos DB storage (aka Polyglot)
 - a. Access the data in Cosmos DB through a user-friendly interface, you will implement a .NET solution that accesses and displays the data from Cosmos DB in a web browser.
 - i. AdventureWorks.Upload
 - ii. AdventureWorks.Web
 - iii. AdventureWorks.Context
 - b. [Exercise: Create Azure Cosmos DB resources by using the Azure portal - Training | Microsoft Learn](#)
 - c. [Work with Azure Cosmos DB - Training | Microsoft Learn](#)

Change Feed

Is Azure Cosmos DB Change Feed a traffic management tool?

Not really. **Change Feed** isn't about routing or balancing traffic like a load balancer or API gateway. Instead, it's a **data event stream** feature. It continuously captures and exposes changes (inserts and updates) in your Cosmos DB container in the order they occur. Think of it as a **real-time log of what changed in your database**, which you can consume to trigger downstream actions.

Why is it useful in the IoT world?

Here's the simple reason:

IoT devices generate **tons of telemetry data**—temperature readings, sensor states, GPS positions, etc. You often need to **react to changes immediately** (e.g., alert when a sensor crosses a threshold, update dashboards, or feed analytics pipelines). Instead of polling the database constantly (which is inefficient and costly), **Change Feed gives you a push-like mechanism**:

- **Real-time processing**: As soon as new data arrives, you can process it.

- **Scalable event-driven architecture:** Hook it up to Azure Functions, Event Hubs, or Stream Analytics for automation.
 - **Reliable ordering:** Events are delivered in the order they were written, which is critical for time-series IoT data.
-

✅ **Example IoT use case:**

Imagine thousands of smart meters sending energy usage data to Cosmos DB. With Change Feed:

- You can trigger an Azure Function whenever usage spikes.
 - Push aggregated data to a dashboard in near real-time.
 - Feed machine learning models for predictive maintenance.
-

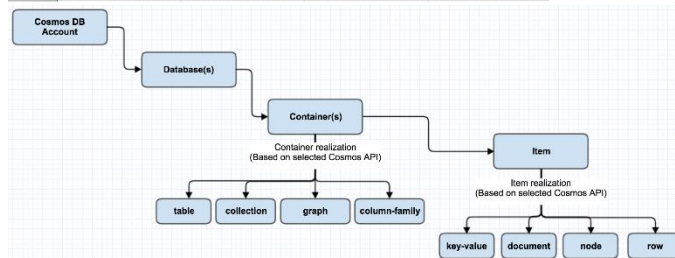
- d. [Working with the change feed - Azure Cosmos DB | Microsoft Learn](#)
 - i. Change feed in Azure Cosmos DB is a persistent record of changes to a container in the order they occur.

How do relationships get created?

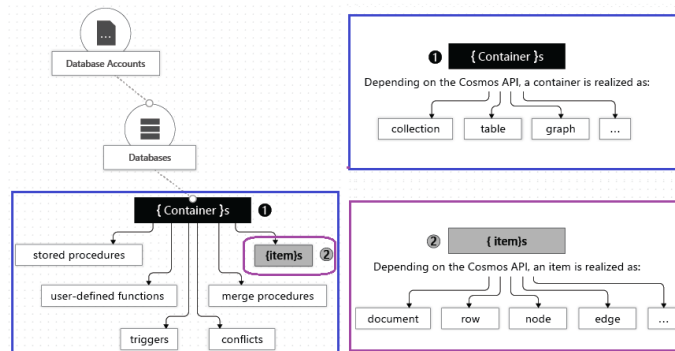
- e. When a container is created, you need to supply a partition key.
 - i. The value of this property is then used to route data to the appropriate partition to be written, updated, or deleted.
 - ii. Items in a container are divided into distinct subsets called logical partitions.
- f. Partitions
 - i. Scale individual containers in a database.
- g. Logical partitions are formed based on the value of a partition key
 - i. All items are grouped by the partition key value in each item in a container.
 - ii. Each item in a container has an item ID (unique within a logical partition).
 - iii. Combining the partition key and the item ID creates the item's index and uniquely identifies the item.
 - iv. The partition key can also be used in the WHERE clause in queries for efficient data retrieval.
 - v. [Partitioning and horizontal scaling - Azure Cosmos DB | Microsoft Learn](#)
- h. [Databases, containers, and items - Azure Cosmos DB | Microsoft Learn](#)

	A	B	C	D
1	PK_ID	fName	mName	lName
2		required	optional	required
3			null	
4				
5		NULL Values		
6				

i.



j.



k.

New Container | Enable Azure Synapse Link

NOSQL API

DATA

ToDoList

Items

Items

Scale & Settings

Stored Procedures

User Defined Functions

Triggers

Conflicts

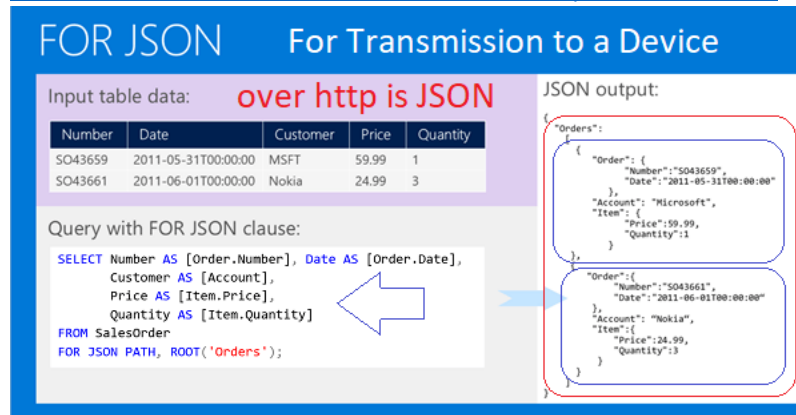
NOTEBOOKS

Notebooks is currently not available. We are working on it.

CosmosDB

l.

- m. [*Quickstart - Create Azure Cosmos DB resources from the Azure portal | Microsoft Learn](#) - simple
- n. [Tutorial: Query data - Azure Cosmos DB for NoSQL | Microsoft Learn](#)
 - i. Uses document and cross product
- o. [NoSQL queries - Azure Cosmos DB for NoSQL | Microsoft Learn](#)
- p. [Work with JSON - Azure Cosmos DB for NoSQL | Microsoft Learn](#)



- q.
- r. [Write stored procedures, triggers, and UDFs in Azure Cosmos DB | Microsoft Learn](#)
 - i. Stored procedures are written using JavaScript, and they can create, update, read, query, and delete items inside an Azure Cosmos DB container.
 - ii. [DP-420: Designing and Implementing Cloud-Native Applications Using Microsoft Azure Cosmos DB \(microsoftlearning.github.io\)](#)
- s. [Tutorial: Create a Jupyter Notebook in Azure Cosmos DB for NoSQL to analyze and visualize data \(preview\) | Microsoft Learn](#)

Home > cosmosdbclasst6



cosmosdbclasst6 | Data Explorer

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Cost Management
- Quick start
- Notifications
- Data Explorer
- Settings

New Container v Enable Azure Synapse Link

NOSQL API

DATA

- ToDoList
 - Scale
 - Items
 - Items
 - Settings
- Stored Procedures
- User Defined Functions
- Triggers
- NOTEBOOKS

Jupyter Notebooks

Notebooks is currently not available. We are working on it.

- t.
- u. Azure Cosmos DB, RU/s stands for Request Units per second. It represents the throughput capacity of a database or container and is a measure of the computational resources (CPU, IO, memory) required to perform database operations. When you provision throughput, you're essentially reserving a specific number of RUs per second for your workload.

Cosmos Workload Type

Microsoft Azure

Home > Azure Cosmos DB >

Create Azure Cosmos DB Account - Azure Cosmos DB for NoSQL

Basics Global distribution Networking Backup Policy Security Tags Review + create

Azure Cosmos DB is a fully managed NoSQL and relational database service for building scalable, high performance applications. Go to production starting at \$24/month per database, multiple containers included. [Learn more](#)

Project Details

Choose a workload type that best aligns with your goals. This helps us provide an optimized starting point for your Azure Cosmos DB account setup. Don't worry—no matter which workload type you choose, you can customize each setting to fit your needs or stick to the defaults provided.

Workload Type * ⓘ

Learning

Best for Learning

Development / Testing

Production

Select the subscription to manage deployed resour

Subscription * AZ-20

Resource Group * Create new

Instance Details

Review + create Previous Next: Global distribution Feedback

V.

options are more about **how you intend to use the account** rather than the specific workload pattern (like IoT or retail catalog).

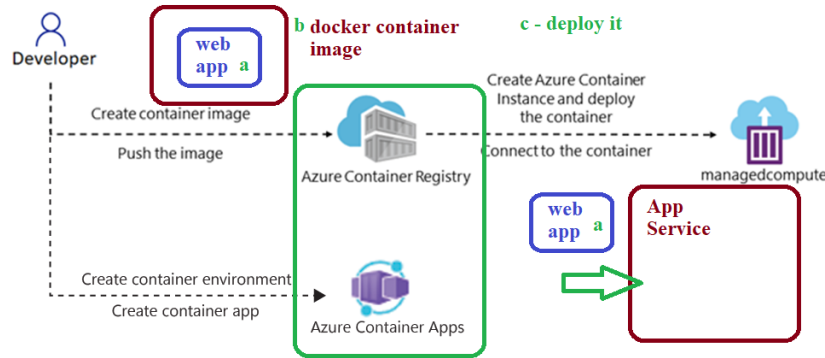
What These Options Actually Do

Option	Purpose
Learning	Ideal for tutorials, quick experiments, or trying out Cosmos DB features. May enable free-tier options or simplified settings.
Development / Testing	Tailored for app development and testing environments. Often disables production-grade features like geo-redundancy to save costs.
Production	Enables high-availability features like multi-region writes and geo-redundancy by default. Designed for live, customer-facing apps.

W.

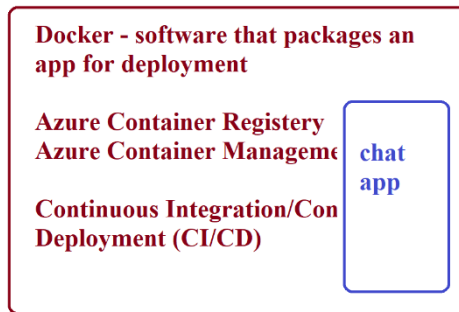
5. Module 05: Implement IaaS solutions

- Console application to display a machine's current IP address.
 - Ipcheck
 - containerapp
- Add **Dockerfile** file to the application so that it could be converted into a Docker container image.
- Deployed the container image to Container Registry.
- [AZ-204: Developing solutions for Microsoft Azure \(microsoftlearning.github.io\)](#)
- [Build a containerized web application with Docker - Training | Microsoft Learn](#)
- [ACR Tasks overview - Azure Container Registry | Microsoft Learn](#)



g.

instead of App Service - Container



deploy from code on a schedule w/
DevOps

h.

6. Module 06: Implement user authentication and authorization

- a. Register an application in Microsoft **Entra ID**, add a user, and then test the user's access to the application to validate that Entra ID can secure it.
 - i. OIDCClient MVC app
- b. You will also use the **Graph Explorer** tool to build and test requests against the Graph API for an Entra ID user account.
- c. [AZ-204: Developing solutions for Microsoft Azure \(microsoftlearning.github.io\)](https://microsoftlearning.github.io/AZ-204: Developing solutions for Microsoft Azure)
- d. [Explore the Microsoft identity platform - Training | Microsoft Learn](#)

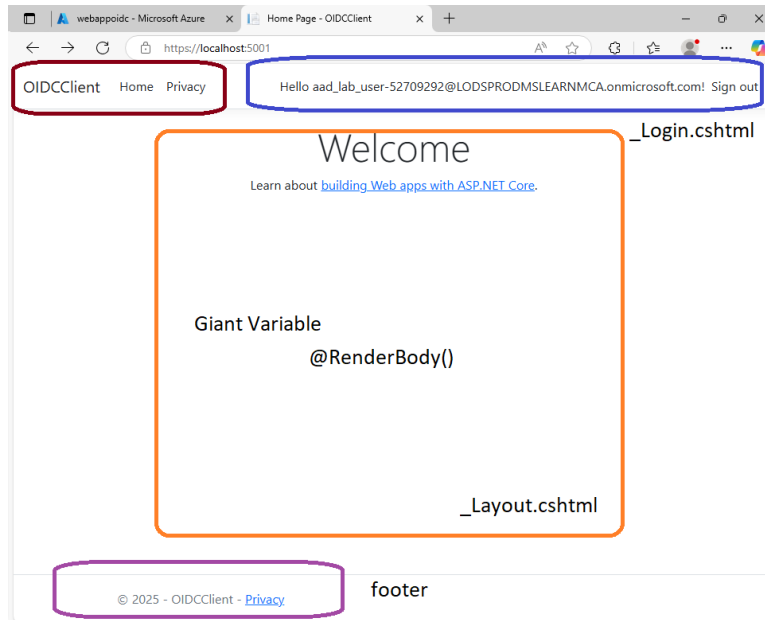
Parameter	Description
/f	Forces deletion of read-only files.
/s	Deletes specified files from the current directory and all subdirectories. Displays the names of the files as they are being deleted.
/q	Specifies quiet mode. You are not prompted for delete confirmation.

e.

- f. [Discover Microsoft Graph - Training | Microsoft Learn](#)

7. Module 07: Implement secure cloud solutions

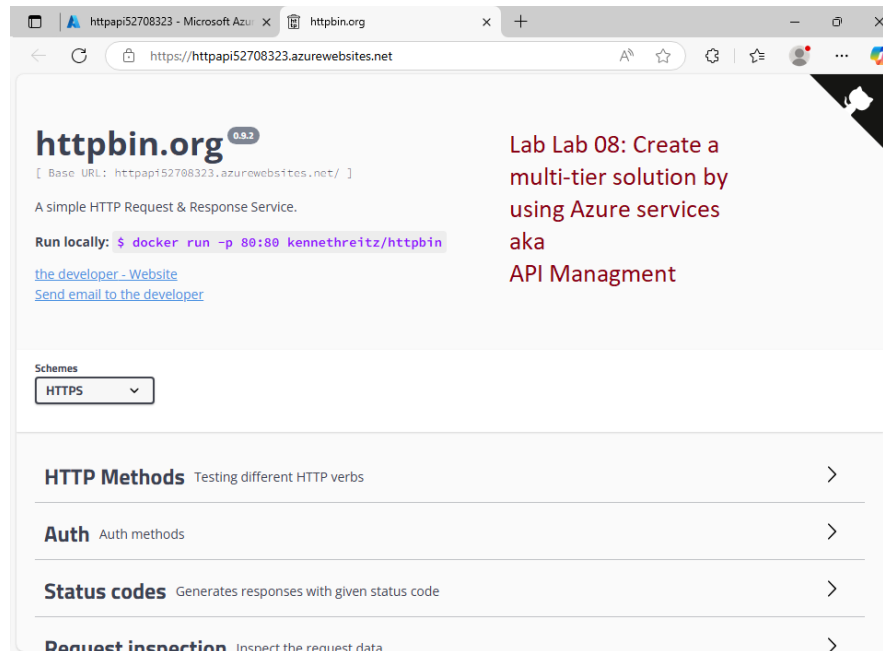
- a. Create a storage account and an Azure Function app that will access the storage account.
 - i. func init
- b. Demonstrate the secure storage of connection string information.
- c. Provision a Key Vault resource and manage the appropriate secrets to store the connection string information.

d. [AZ-204: Developing solutions for Microsoft Azure \(microsoftlearning.github.io\)](https://microsoftlearning.github.io/AZ-204: Developing solutions for Microsoft Azure)

e.

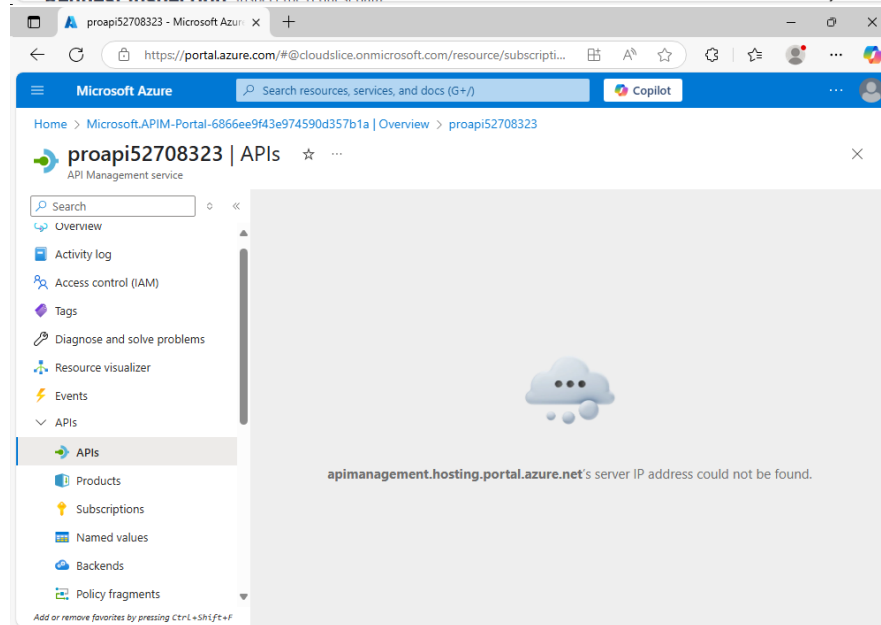
8. Module 08: Implement API Management

- a. Define a new API
- b. Create a containerized (Docker) application to host a web app on Azure, as the source of information for the API.
 - i. kennethreitz/httpbin:latest
 - ii. [kennethreitz/httpbin - Docker Image | Docker Hub](https://hub.docker.com/r/kennethreitz/httpbin)
 - iii. Httpbin.org
 - iv. [Releases · Kenneth Reitz / httpbin · GitLab](https://github.com/kennethreitz/httpbin/releases)
- c. You will then build an API proxy using the Azure API Management capabilities to expose and test your APIs.
- d.

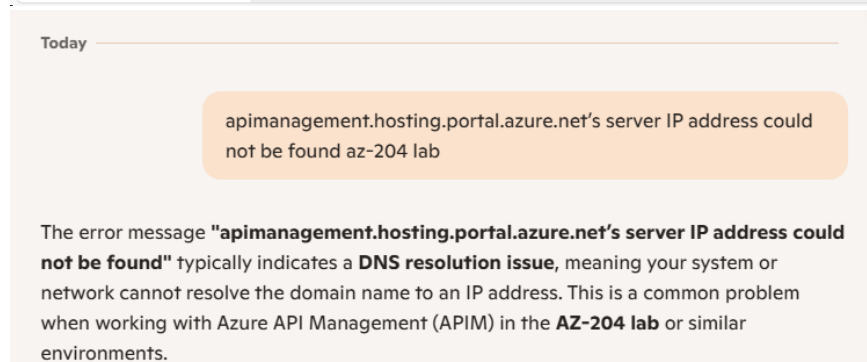


Lab Lab 08: Create a multi-tier solution by using Azure services aka API Management

e.



f.



g.

APPLIES TO: All API Management tiers

- h. In Azure API Management, a product contains one or more APIs, a usage quota, and the terms of use. After a product is published, developers can subscribe to the product and begin to use the product's APIs.

i. https://learn.microsoft.com/en-us/azure/api-management/api-management-howto-add-products?WT.mc_id=Portal-fx&tabs=azure-portal&pivots=interactive

♦ **What the Forward Slash / Denotes in a URL**

Position in URL	Meaning
After the domain (e.g., <code>https://example.com/</code>)	Denotes the root of the website or web application
Between path segments (e.g., <code>/products/shoes</code>)	Separates hierarchical levels in the path
At the end of a path (e.g., <code>/about/</code>)	Often denotes a directory rather than a file (though this depends on server configuration)
Double slashes after protocol (e.g., <code>https://</code>)	Separates the protocol from the rest of the URL
In query strings or fragments	Rarely used here; more common delimiters are <code>?</code> , <code>&</code> , and <code>#</code>

j.

🔗 **Examples**

- `https://example.com/` → The root of the site.
- `https://example.com/blog/` → The `/blog/` directory or route.
- `https://example.com/blog/post-1` → A specific resource under `/blog/`.

✳ **Special Notes**

- A single `/` by itself in a GET request (like `GET / HTTP/1.1`) refers to the **root path** of the server.
- In RESTful APIs, slashes define **resource hierarchies**:
 - `GET /users/123/orders` → Get orders for user with ID 123.

k.

🎨 **Frontend vs Backend Testing: What They Show**

Aspect	Frontend Testing	Backend Testing
What it tests	User interface (UI), layout, buttons, forms, responsiveness	Server logic, APIs, databases, authentication, business rules
Who sees it	End users (what they interact with)	Developers and systems (behind the scenes)
Tools used	Selenium, Cypress, Playwright	Postman, JMeter, SQL tools
Focus	Usability, visual correctness, cross-browser/device compatibility	Data integrity, performance, security, logic correctness
Example	Does the "Submit" button work and look right?	Does the form actually save data to the database correctly?

l.

Stage	Direction	Purpose	Example Use Case
Inbound	Request	Preprocess client request	Add auth headers, validate tokens
Backend	Request	Configure backend call	Change backend URL, set timeout
Outbound	Response	Postprocess backend response	Format response, redact data

m.

the terms **inbound**, **backend**, and **outbound** refer to different stages of the **API request/response pipeline**. Understanding these stages helps you apply the right logic or transformation at the right time.

The API Request Lifecycle

When a client sends a request to an API through APIM, it flows through these stages:

Client → Inbound → Backend → Outbound → Client

n.

◆ Inbound

- **What it is:** The stage where the request **enters** the API gateway from the client.
- **Purpose:** Modify or inspect the request **before** it reaches the backend.
- **Common uses:**
 - Authentication and authorization
 - Rate limiting
 - Header or query parameter transformation
 - Logging or telemetry

Example:

```
Xml Copy
<inbound>
  <set-header name="x-api-key" exists-action="override">
    <value>my-secret-key</value>
  </set-header>
</inbound>
```

o.

◆ Backend

- **What it is:** The stage where the request is **prepared to be sent** to the actual backend service.
- **Purpose:** Configure how the gateway communicates with the backend.
- **Common uses:**
 - Set timeout or retry policies
 - Change the backend URL dynamically
 - Add backend-specific headers

Example:

```
Xml Copy
<backend>
  <set-backend-service base-url="https://my-backend.com/api" />
</backend>
```

p.

◆ Outbound

- **What it is:** The stage where the **response from the backend** is processed **before** it's sent back to the client.
- **Purpose:** Modify or enrich the response.
- **Common uses:**
 - Format transformation (e.g., XML to JSON)
 - Remove sensitive headers
 - Add caching headers
 - Mask or redact data

Example:

```

Xml
Copy

<outbound>
  <set-header name="Cache-Control" exists-action="override">
    <value>no-store</value>
  </set-header>
</outbound>

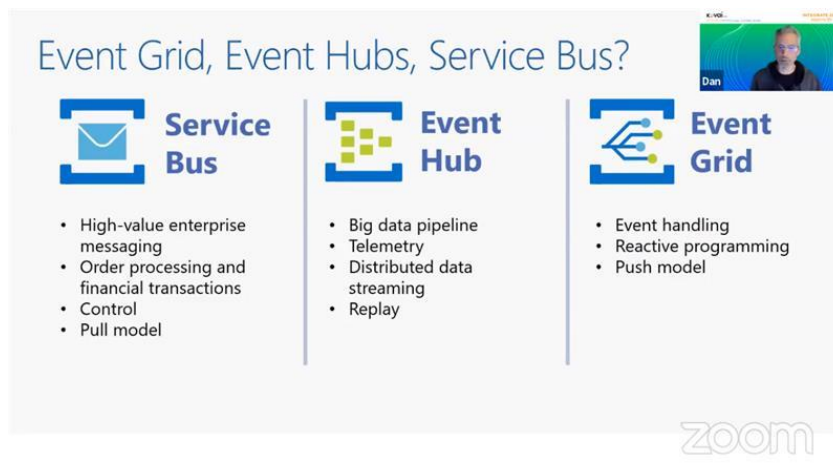
```

q.

r. [AZ-204: Developing solutions for Microsoft Azure \(microsoftlearning.github.io\)](https://microsoftlearning.github.io/AZ-204: Developing solutions for Microsoft Azure)

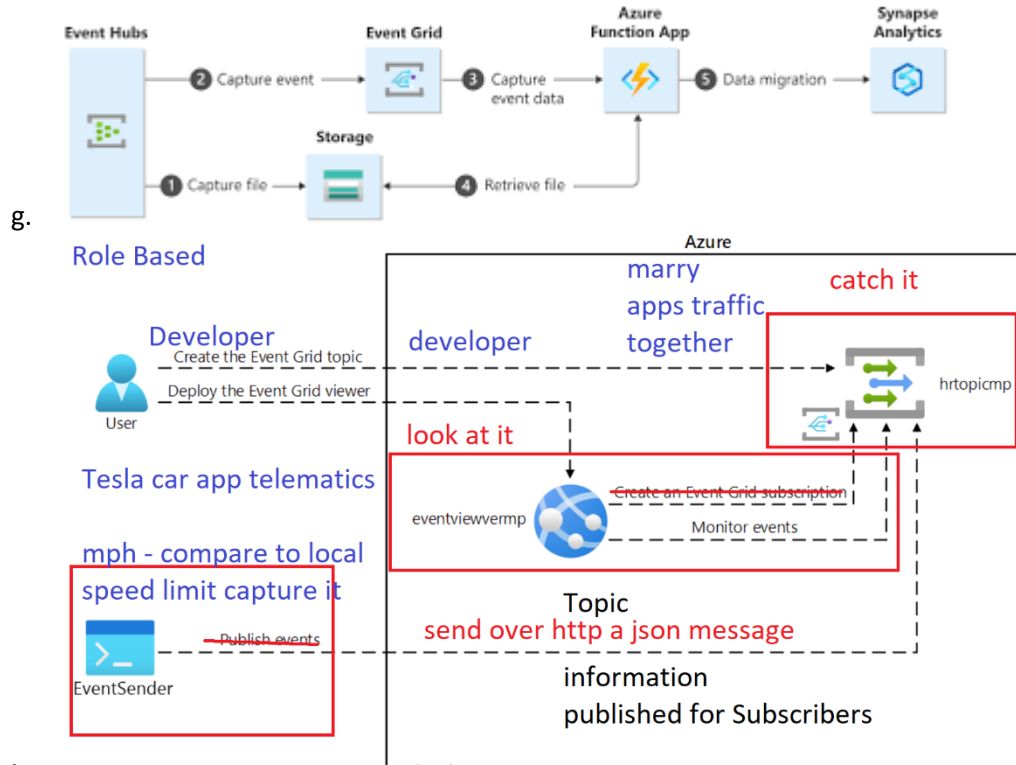
9. Module 09: Develop event-based solutions

- [Azure messaging services documentation | Microsoft Learn](#)
 - Event Grid pushes the ingested data to the subscribers whereas Event Hubs makes the data available in a pull model
- [Overview - Azure Event Grid | Microsoft Learn](#)
- Start with a proof-of-concept web app, hosted in a container, that will be used to subscribe to your **Event Grid**.
 - microsoftlearning/azure-event-grid-viewer:latest
 - EventPublisher console
- This app will allow you to submit events and receive confirmation messages that the events were successful.



e.

f. [Tutorial: Send Event Hubs data to data warehouse - Event Grid - Azure Event Grid | Azure Docs](#)



- h.
- i. [Use cases for using Azure Event Grid - Azure Event Grid | Microsoft Learn](#)
- j. [AZ-204: Developing solutions for Microsoft Azure \(microsoftlearning.github.io\)](#)
- k. [Explore Azure Event Grid - Training | Microsoft Learn](#)
- l. [Azure Event Hubs – data streaming platform with Kafka support - Azure Event Hubs | Microsoft Learn](#)
- m. [Exercise: Send and receive message from a Service Bus queue by using .NET. - Training | Microsoft Learn](#)
- n. [Quickstart: Send custom events to Azure Function - Event Grid - Azure Event Grid | Microsoft Learn](#)
- o. [Azure Event Hubs – data streaming platform with Kafka support - Azure Event Hubs | Microsoft Learn](#)
 - i. Azure Event Hubs is a cloud native data streaming service that can stream millions of events per second
 - ii. A key difference between Event Grid and Event Hubs is in the way event data is made available to the subscribers.
 - iii. Event Grid pushes the ingested data to the subscribers
 - iv. Event Hubs makes the data available in a pull model. As events are received, Event Hubs appends them to the stream.
 - v. [Asynchronous messaging options - Azure Architecture Center | Microsoft Learn](#)
- p. Event Grid Topics
 - i. **Send events to subscribers (push delivery)** or subscribers can connect to Event Grid to read events (pull delivery).

- ii. MQTT, which stands for Message Queuing Telemetry Transport, is a lightweight, open, and simple messaging protocol designed for constrained devices and low-bandwidth, high-latency, or unreliable networks.
- iii. It operates on a **publish-subscribe model, where devices (publishers) send messages to a central broker, and other devices (subscribers) receive those messages based on their subscriptions.**
- iv. MQTT is widely used in the Internet of Things (IoT) for connecting devices and enabling communication between them.



MQTT vs HTTP: Protocol Comparison

Feature	MQTT	HTTP
Type	Message-based publish/subscribe	Request/response
Transport	TCP (often on port 1883), or WebSockets	TCP (usually over port 80/443)
Communication Model	Asynchronous, many-to-many (pub/sub)	Synchronous, one-to-one (client-server)
Overhead	Very low (ideal for constrained devices)	Higher (verbose headers, full HTTP stack)
Text/Binary	Binary protocol	Text-based (headers, body, etc.)
Data Format	Arbitrary (e.g., JSON, binary blobs)	Typically HTML/JSON/XML
Statefulness	Long-lived connections (persistent sessions)	Stateless (each request is new)
Use Case Fit	IoT, telemetry, sensors, embedded devices	Web browsing, APIs, standard clients

- v. [Azure-Samples/MqttApplicationSamples: Samples implementing common PubSub patterns for Edge and Cloud Brokers](#)
- vi. [Create, view, and manage Azure Event Grid namespaces - Azure Event Grid | Microsoft Learn](#)
- vii. [Create, view, and manage Azure Event Grid namespaces - Azure Event Grid | Microsoft Learn](#)
- viii. A namespace in Azure Event Grid is a logical container for one or more topics, clients, client groups, topic spaces, and permission bindings.
- ix. A Topic is a logical channel or name used to group events.
- x. Think of it as a "category" or "stream" where events are published, and subscribers can listen.
- xi. Acts as a routing address for published events..
- xii. Allows event publishers to send messages to a known location
- xiii. The event subscription is what connects a handler (like a function or webhook) to a specific topic.
- xiv. Subscribers then attach to that topic to receive only the events published to it.
- xv. Schema Format

First: Yes, **events are always JSON-formatted**

But... the **structure** of the JSON changes depending on the **schema** you choose.

The 3 Event Schemas in Azure Event Grid

Schema	Structure	Who defines it?	Best used when...
Event Grid Schema	Native to Azure	Microsoft	You're working within Azure's ecosystem
CloudEvents v1.0	CNCF standard	CloudEvents community	You want portability across clouds/systems
Custom Schema	You define it	You	You have legacy or domain-specific formats

xvi.

1. Event Grid Schema (Default)

Azure-native format. Fields include:

```

{
  "id": "123",
  "eventType": "Contoso.Items.ItemReceived",
  "subject": "/items/blue-widget",
  "eventTime": "2025-07-07T18:30:52.123Z",
  "data": {
    "itemSku": "SKU12345"
  },
  "dataVersion": "1.0",
  "metadataVersion": "1"
}

```

2. CloudEvents v1.0

Industry standard schema — useful for cross-cloud or Kubernetes environments:

```

{
  "specversion": "1.0",
  "type": "com.contoso.items.received",
  "source": "/items",
  "id": "123",
  "time": "2025-07-07T18:30:52.123Z",
  "datacontenttype": "application/json",
  "data": {
    "itemSku": "SKU12345"
  }
}

```

xvii.

q. Event Grid Viewer

The Azure Event Grid Viewer is a **sample web app** (typically hosted on Azure App Service) that:

- Displays incoming **event messages** from an Event Grid topic or domain
- Is often used to **visualize or debug** the event flow
- Is **read-only** — it does not send or generate events, just listens

i.



How it works

1. You deploy the Event Grid Viewer (a basic web app).
2. You create an **event subscription** with the **viewer URL** as the endpoint.
3. It listens and renders incoming event data (as JSON in the UI).

ii.

10. Module 10: Develop message-based solutions

- a. Create a proof of concept for this scenario by employing an **Azure Service Bus Queue**.
- b. Create a .NET Core project that will publish messages to the system,
- c. A second .NET Core application that will read messages from the queue.
- d. [*Discover Azure message queues - Training | Microsoft Learn](#)
 - i. Service Bus, Azure Queue Storage, messages
- e. [AZ-204: Developing solutions for Microsoft Azure \(microsoftlearning.github.io\)](#)
- f. MessageReader console
- g. MessagePublisher console

11. Module 11: Monitor and optimize Azure solutions

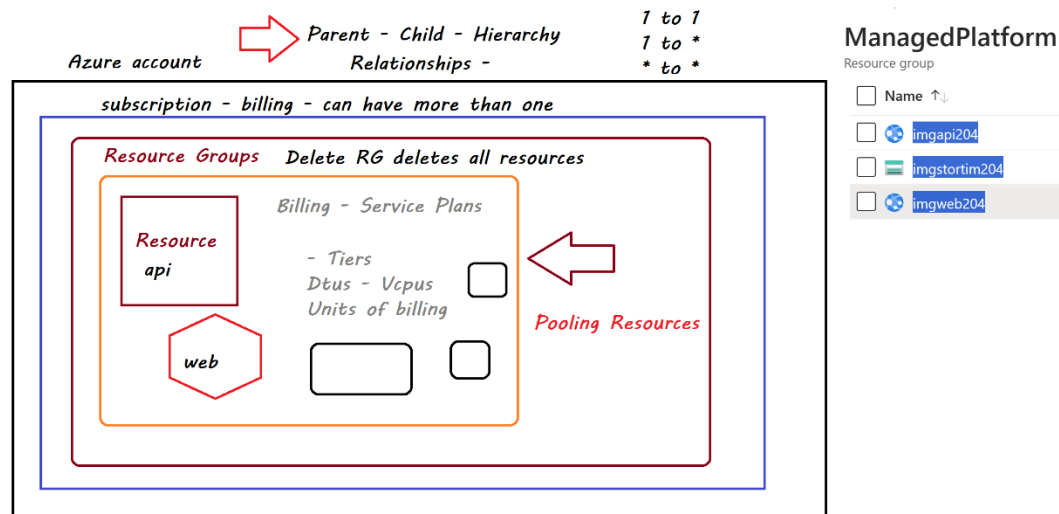
- a. [Application Insights OpenTelemetry observability overview - Azure Monitor | Microsoft Learn](#) (Azure Monitor)

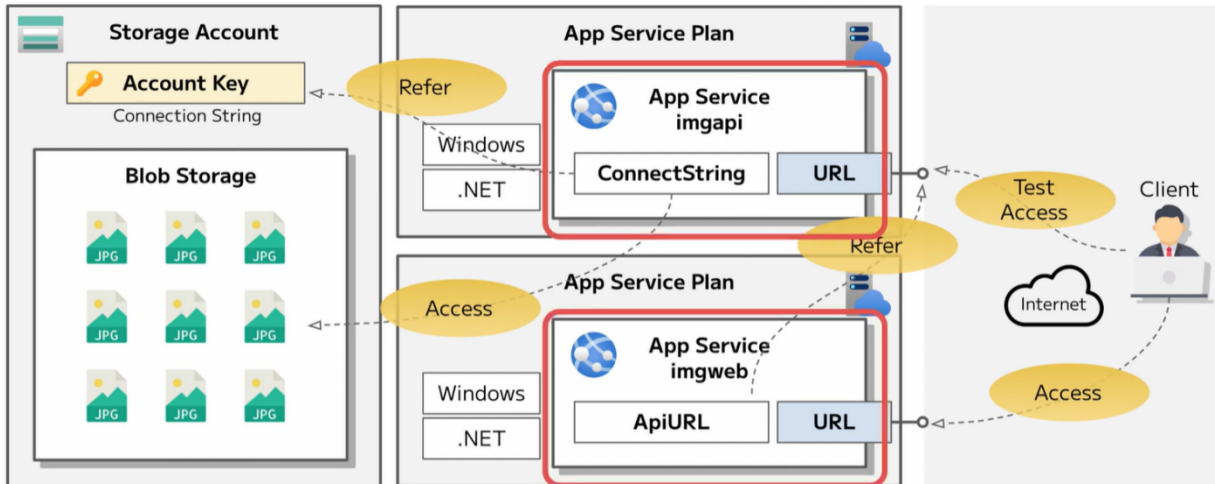
- b. Azure Application Insights is an extensible Application Performance Management (APM) service provided by Azure Monitor.
 - c. It allows developers and DevOps teams to monitor the performance and usage of their live web applications, regardless of where they are hosted (cloud or on-premises).
 - d. It provides insights into application behavior, identifies performance bottlenecks, and helps diagnose errors.
 - e. [Troubleshoot solutions by using Application Insights - Training | Microsoft Learn](#)
 - f. [Design to scale out - Azure Architecture Center | Microsoft Learn](#)
 - g. Create an Application Insights resource in Azure that will be used to monitor and log application insight data for later review.
 - i. smpapi
 - ii. instrm
 - iii. SimpleApi
12. Module 12: Integrate caching and content delivery within solutions
- a. implement the Azure Content Delivery Network capabilities to provide a caching solution based on customer locations.
 - b. The lab configures a storage account for image and video files, which are impacted the most by the latency issues.
13. End list

[GitHub - MicrosoftLearning/AZ-204-DevelopingSolutionsforMicrosoftAzure: AZ-204: Developing solutions for Microsoft Azure](#)

MS Learn Instructions for Labs: Just TOC no code.

[AZ-204: Developing solutions for Microsoft Azure \(microsoftlearning.github.io\)](#)





Module	Lab
1. Learning Path 01: Implement Azure App Service Web Apps	2. Lab 01: Build a web application on Azure platform as a service offerings
3. Learning Path 02: Implement Azure Functions	4. Lab 02: Implement task processing logic by using Azure Functions
5. Learning Path 03: Develop solutions that use blob storage	6. Lab 03: Retrieve Azure Storage resources and metadata by using the Azure Storage SDK for .NET
7. Learning Path 04: Develop solutions that use Cosmos DB storage	8. Lab 04: Construct a polyglot data solution
9. Learning Path 05: Implement containerized solutions	10. Lab 05: Deploy compute workloads by using images and containers
11. Learning Path 06: Implement user authentication and authorization	12. Lab 06: Authenticate by using OpenID Connect, MSAL, and .NET SDKs
13. Learning Path 07: Implement secure Azure solutions	14. Lab 07: Access resource secrets more securely across services

Module	Lab
15. Learning Path 08: Implement API Management	16. Lab 08: Create a multi-tier solution by using Azure services
17. Learning Path 09: Develop event-based solutions	18. Lab 09: Publish and subscribe to Event Grid events
19. Learning Path 10: Develop message-based solutions	20. Lab 10: Asynchronously process messages by using Azure Service Bus Queues
21. Learning Path 11: Troubleshoot solutions by using Application Insights	22. Lab 11: Monitor services that are deployed to Azure
23. Learning Path 12: Implement caching for solutions	24. Lab 12: Enhance a web application by using the Azure Content Delivery Network

PowerShell on Local Windows

1. [Install Azure PowerShell on Windows | Microsoft Learn](#)

```
$PSVersionTable.PSVersion
```

Major	Minor	Build	Revision
5	1	19041	3031

windows vs VS code

Same PC



Major	Minor	Patch	PreReleaseLabel	BuildLabel
7	3	7		

- 2.
3. Multiple versions of PowerShell – Inside VS Code vs. Win Terminal
4. Update-Module -Name Az -Force
5. [HTTP response status codes - HTTP | MDN \(mozilla.org\)](#)
6. 202: Accepted - indicates that server has received and understood the request, and that it has been accepted for processing
7. 500 : Internal Server error - indicates that the server encountered something it didn't expect and was unable to complete the request.
8. Static \$web site in Storage account

Home > imgstortim204 | Containers >

\$web Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Shared access tokens

Access policy

Properties

Metadata

Editor (preview)

static website in a storage acct.**Change access level**

Change the access level of container '\$web'.

Anonymous access level

Private (no anonymous access)

Anonymous access to this container is being blocked because anonymous access is disabled on this storage account.

OK

Cancel

This XML file does not appear to have any style information associated with it. The document tree is shown below.

message

```
<Error>
  <Code>PublicAccessNotPermitted</Code>
  <Message>Public access is not permitted on this storage account.
  RequestId:a3b7b9f4-e01e-0079-1fb2-fc4323000000 Time:2023-10-
  12T02:17:33.7042886Z</Message>
</Error>
```

9.

Microsoft Azure Search resources, services, and docs (G+)

Home >

imgstortim204 Storage account

Search

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Tags (edit)

Add tags

Properties

Monitoring

Capabilities (7)

Recommendations

Blob service

Hierarchical namespace Disabled

Default access tier Hot

Blob anonymous access Disabled

10.

Microsoft Azure Search resources, services, and docs (G+)

Home > imgstortim204

imgstortim204 | Static website Storage account

Search

Save Discard Give feedback

Static website

Static website Disabled Enabled

Enabling static websites on the blob service allows you to host static content. Webpages may include static content and client-side scripts. Server-side scripting is not supported. As data is replicated asynchronously from primary to secondary regions, files at the secondary endpoint may not be immediately available or in sync with files at the primary endpoint. [Learn more](#)

11.

enable anonymous access - see value in Overview page also

Home > imgstortim204

imgstortim204 | Configuration   ...

Storage account

Search << Save Discard Refresh Give feedback

Settings

- Configuration
- Data Lake Gen2 upgrade
- Resource sharing (CORS)
- Advisor recommendations
- Endpoints
- Locks

Monitoring

- Insights
- Alerts
- Metrics

The cost of your storage account depends on the usage and the options you choose below.

Account kind
StorageV2 (general purpose v2)

Performance ⓘ
☒ Standard ☐ Premium

ⓘ This setting cannot be changed after the storage account is created.

Secure transfer required ⓘ
☐ Disabled ☒ Enabled

Allow Blob anonymous access ⓘ
☒ Disabled ☐ Enabled

Allow storage account key access ⓘ
☐ Disabled ☒ Enabled

12.

Home > imgstortim204 | Containers >

\$web ...

Container

Search << Upload Change access level Refresh Delete Chan

Overview

- Diagnose and solve problems
- Access Control (IAM)

Settings

- Shared access tokens
- Access policy
- Properties
- Metadata
- Editor (preview)

Enable access - Config blade and Container Access Level - Static Web Site Storage

Change access level
Change the access level of container '\$web'.

Anonymous access level ⓘ
Blob (anonymous read access for blobs only) ▼

⚠ Blobs within the container can be read by anonymous request, but container data is not available. Anonymous clients cannot enumerate the blobs within the container.

OK Cancel

13.

Data Movement - Storage

1. What are the boundaries – what makes up an “item” – Query it...
2. Question (report – data model – context – my object is a ?): How many bicycles are sold in Jan 2022 – Green in color – Size 26” wheel – boys – w/baskets
3. Data Relations – give data meaning: 1 to 1, 1 to many, many to many, many to 1

one user with many orders

Relationships in data

Shopping cart

**product
price
tax**

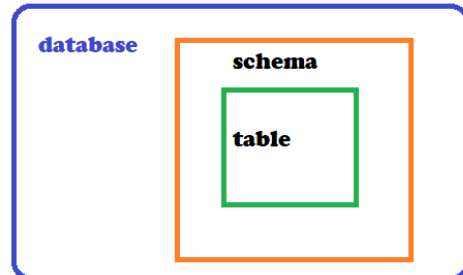
**Store data or
information**

shipping info

**Unstructured
no relationships
to other
information**

user profile info

array · no json ?



log file - my_log.ldf
track of change

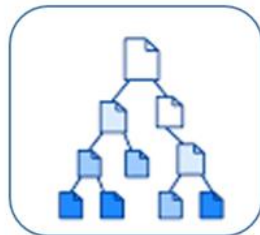
data in



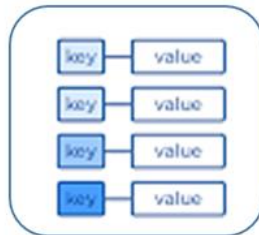
main data file mydatabase.mdf
store data

4.

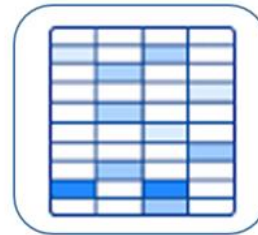
did i get the freshest data or an old copy?



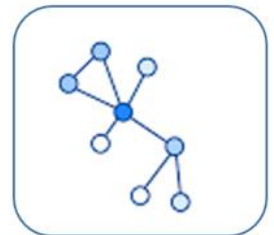
Document
Store



Key-Value
Store

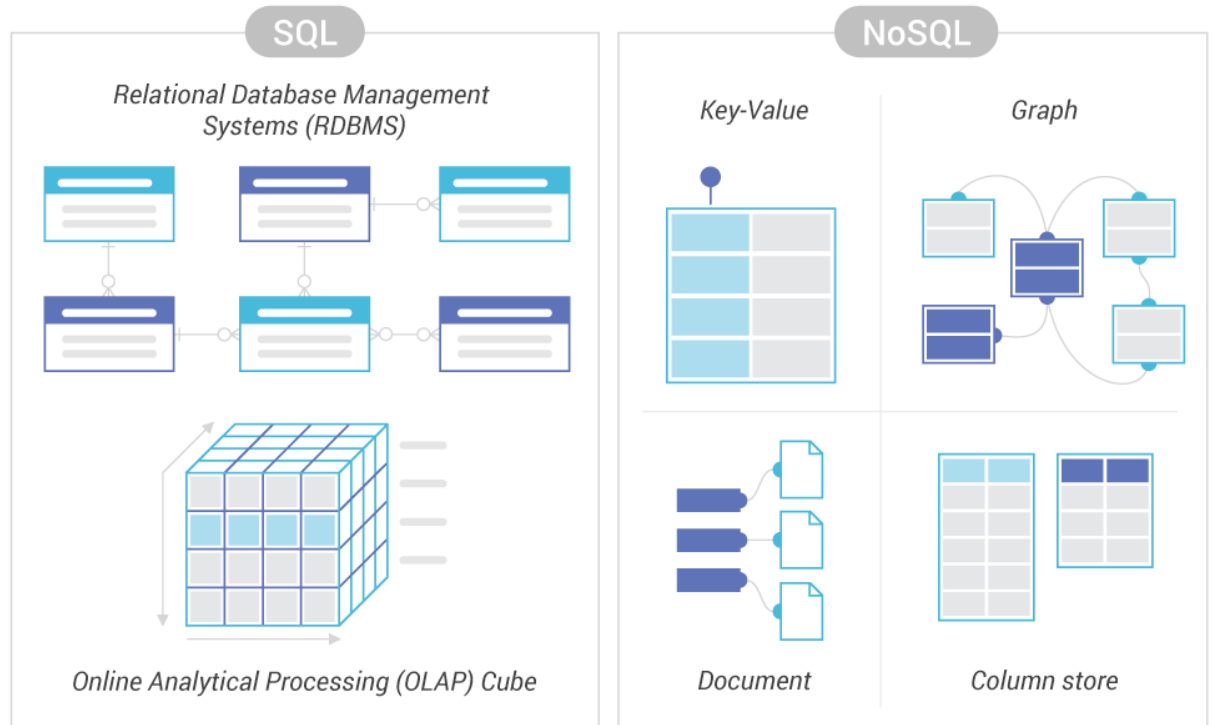


Wide-Column
Store



Graph
Store

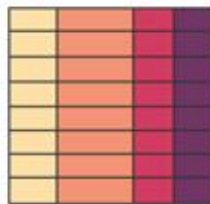
5.



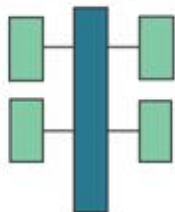
6.



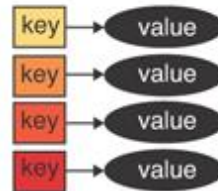
Relational



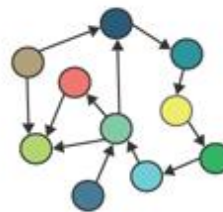
Analytical (OLAP)



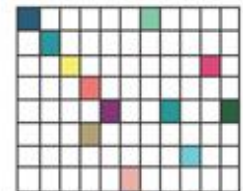
Key-Value



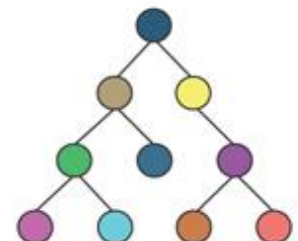
Graph



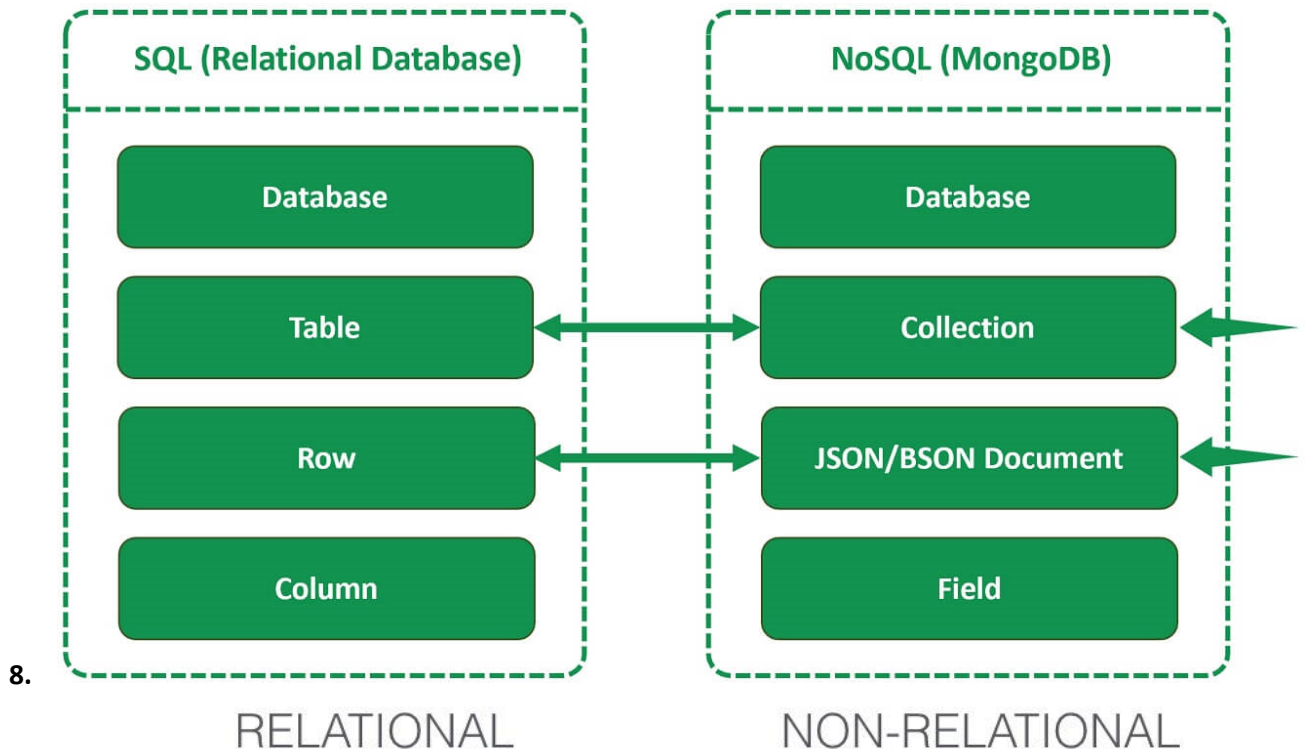
Column-Family



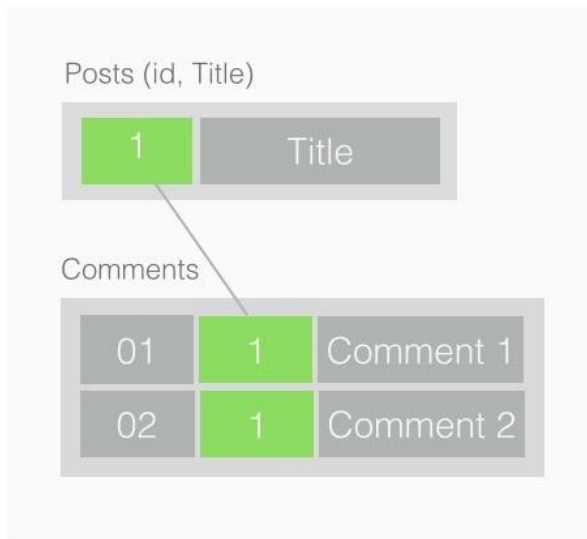
Document

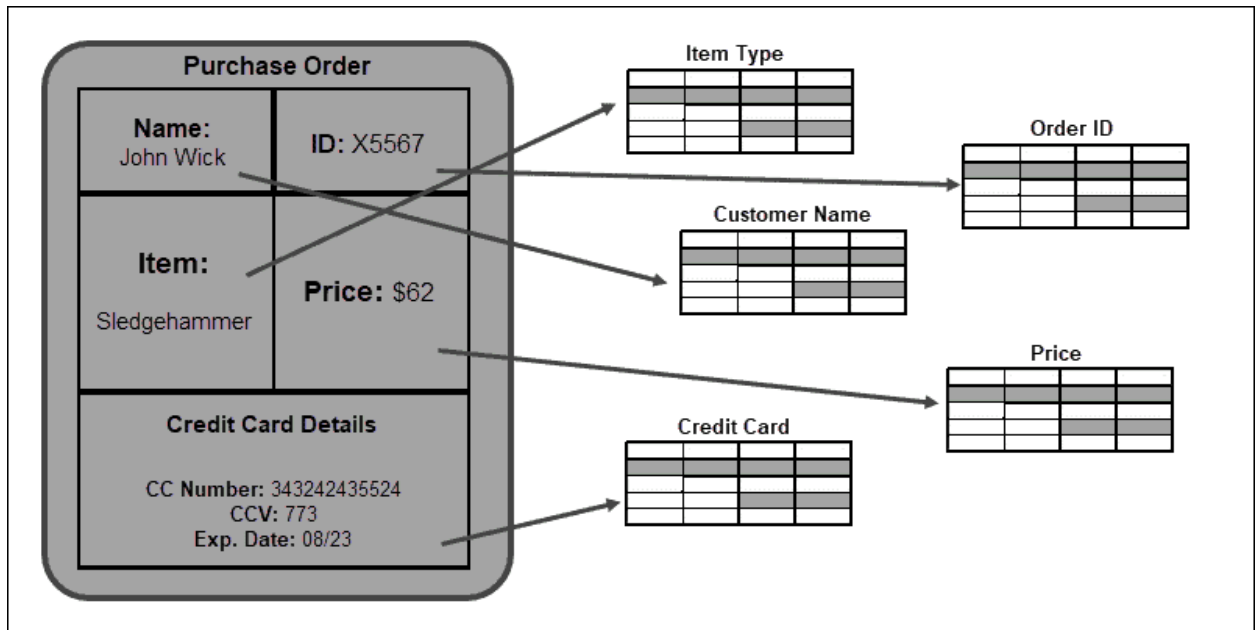


7.



- 9.
- 10.
- 11.
- 12.

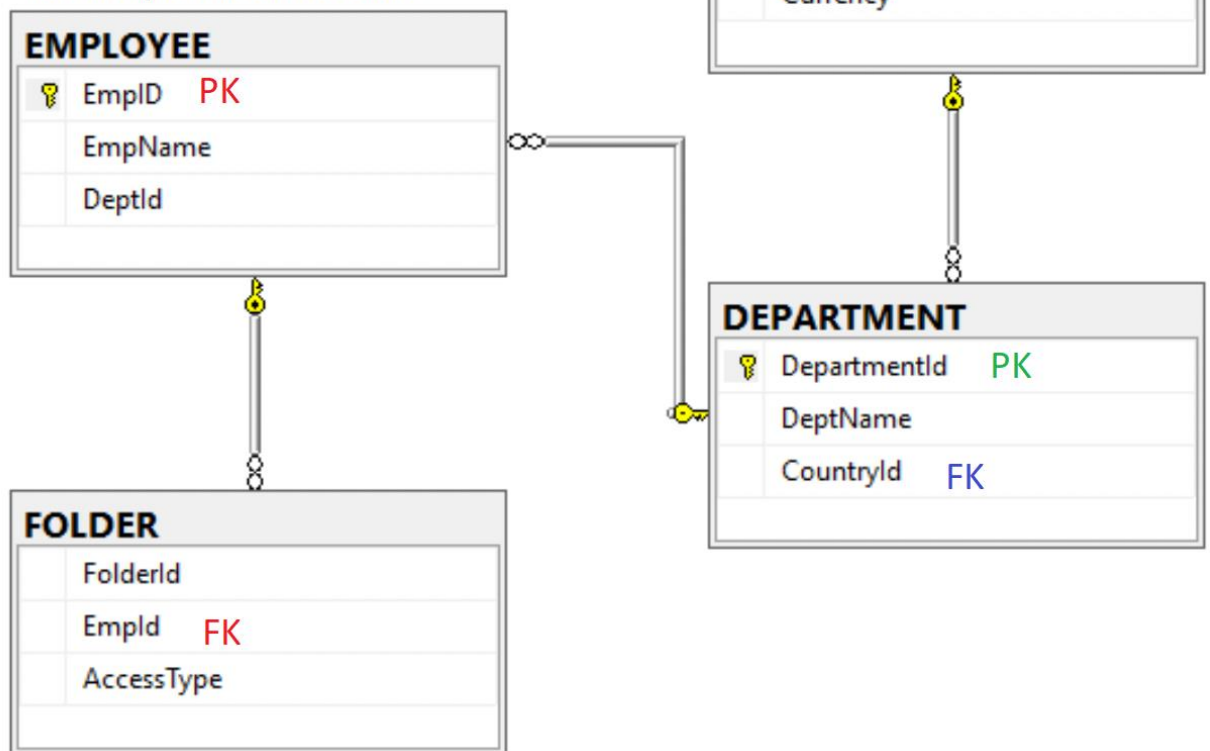




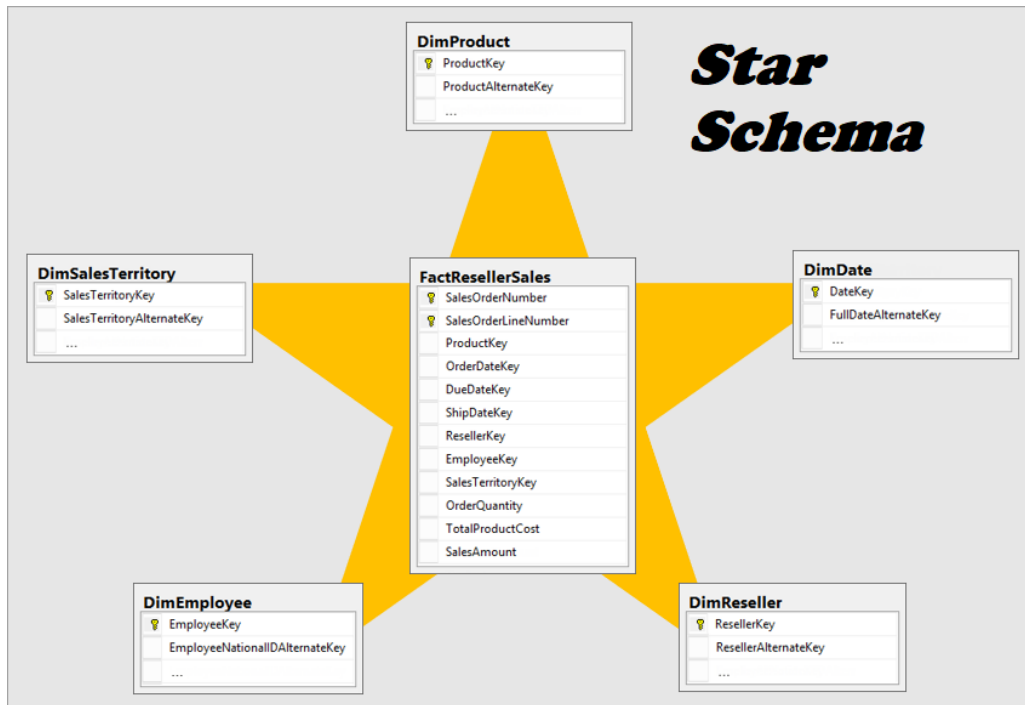
13.

PK sorts the table and gives a record ID for each row in each table

FK map one table to another to identify related records



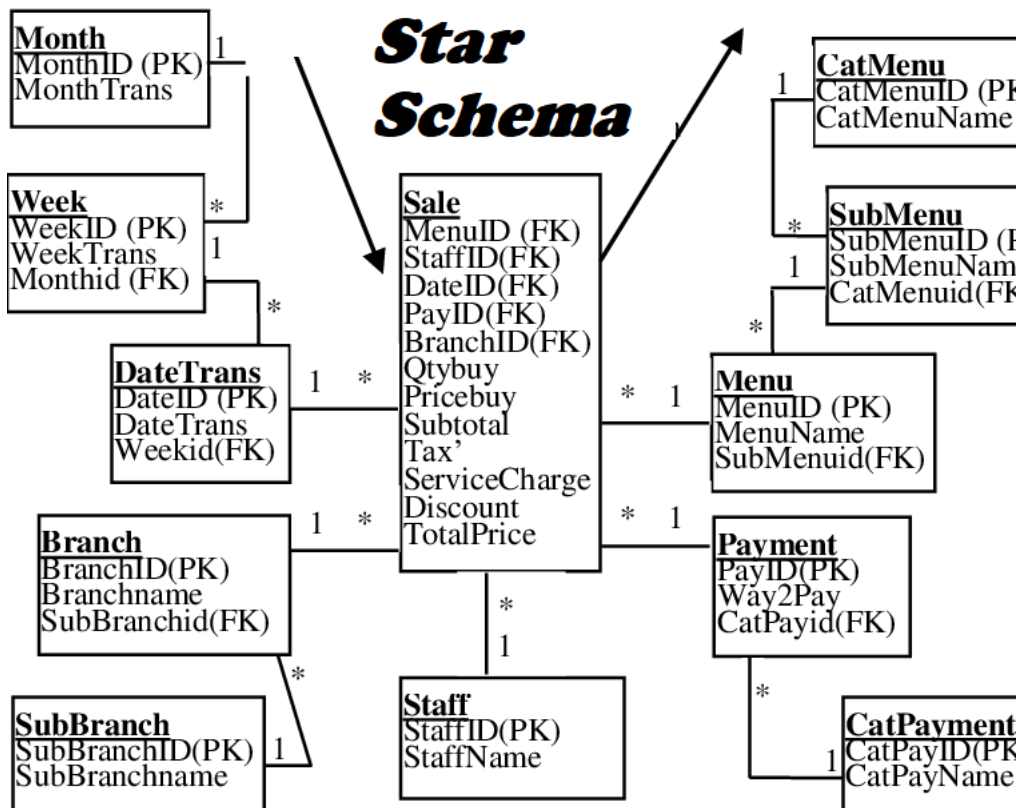
14.



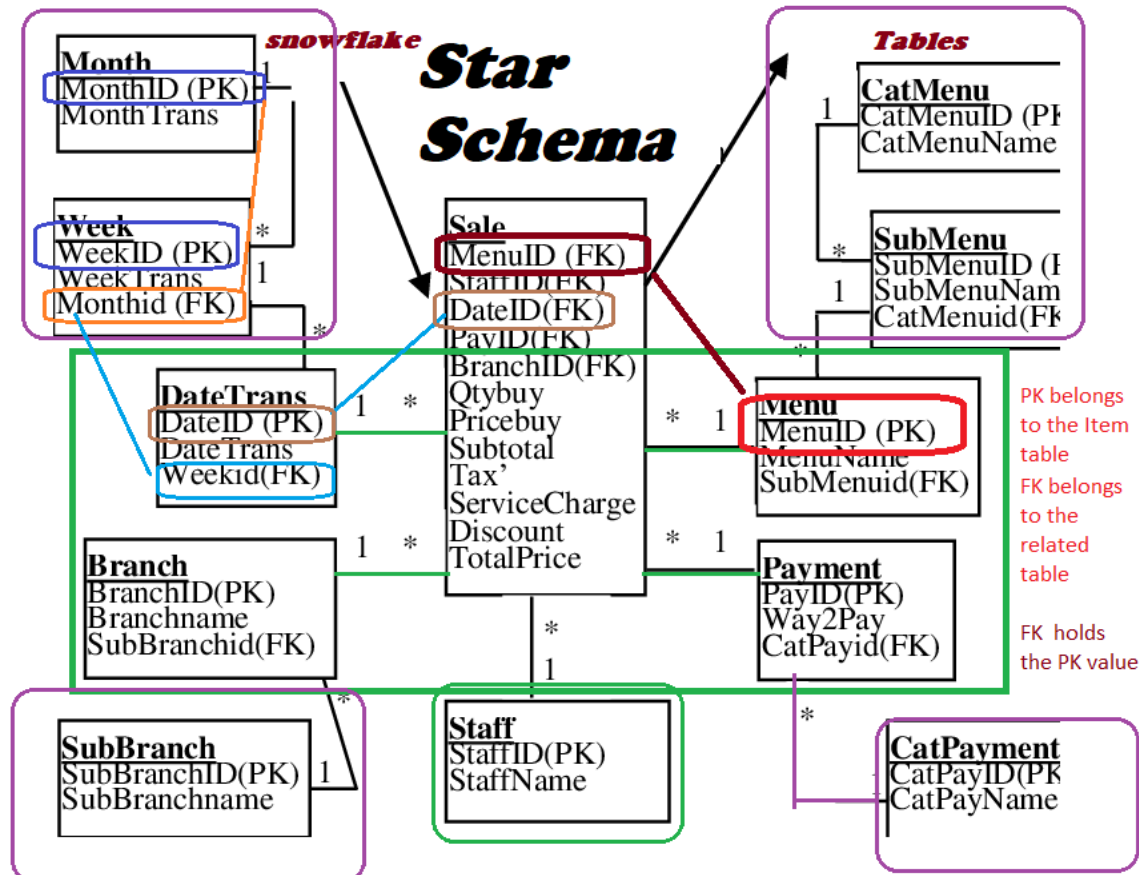
15.

16.

17.



18.

fact table - holds numbers to analyze**Warehouse****dimension tables filter - slice - data**

19.

Table also called Relation

Primary Key Domain
Ex: NOT NULL

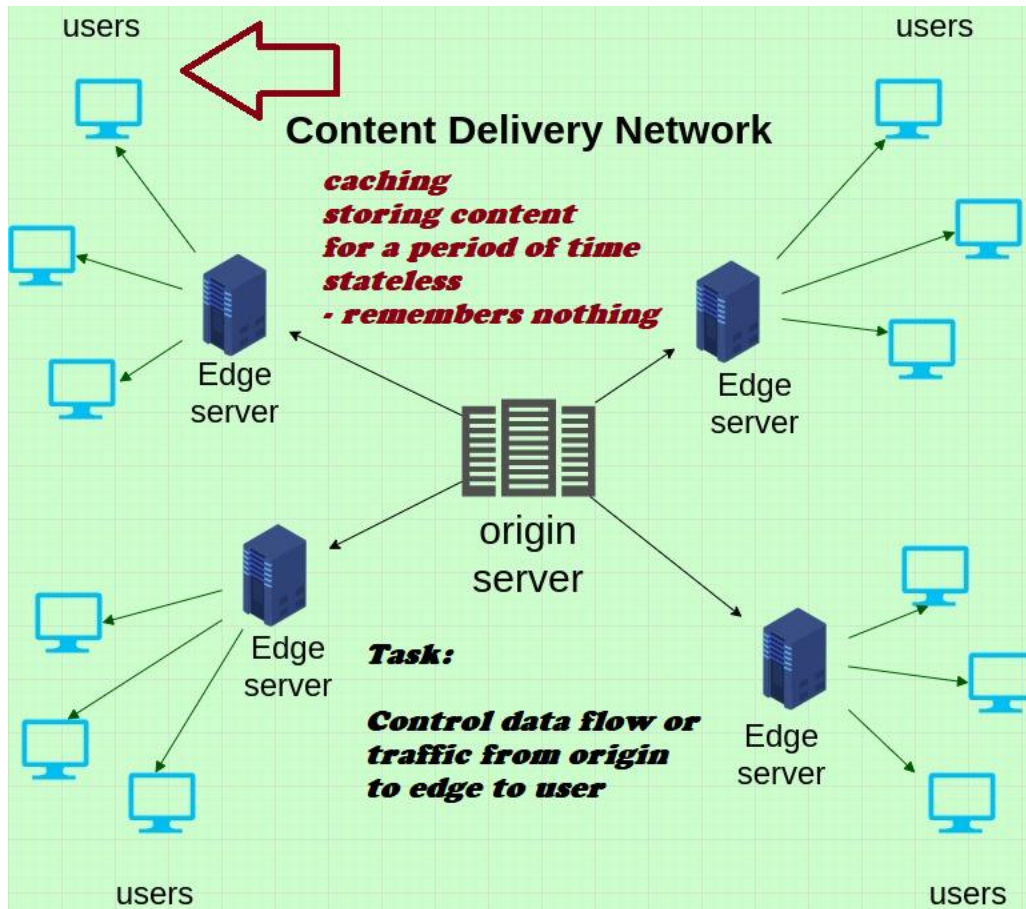
© guru99.com

CustomerID	CustomerName	Status
1	Google	Active
2	Amazon	Active
3	Apple	Inactive

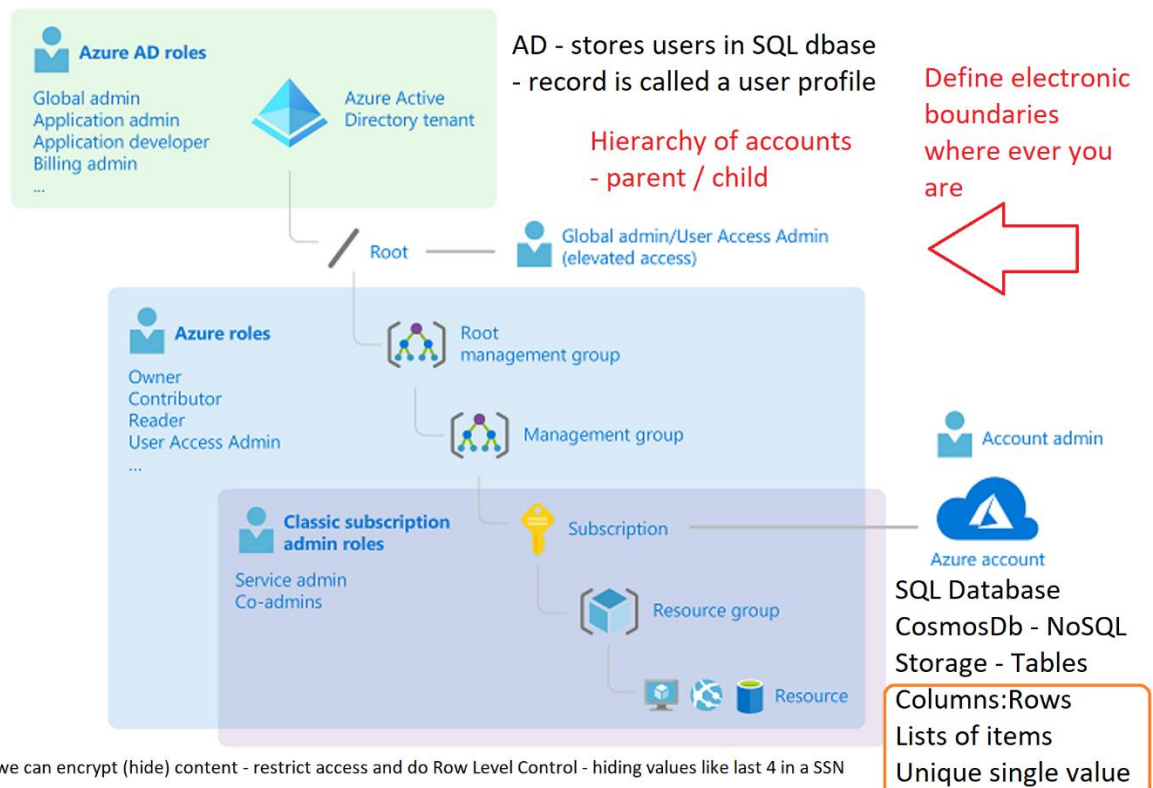
Tuple OR Row
Total # of rows is **Cardinality**

Column OR Attributes
Total # of column is **Degree**

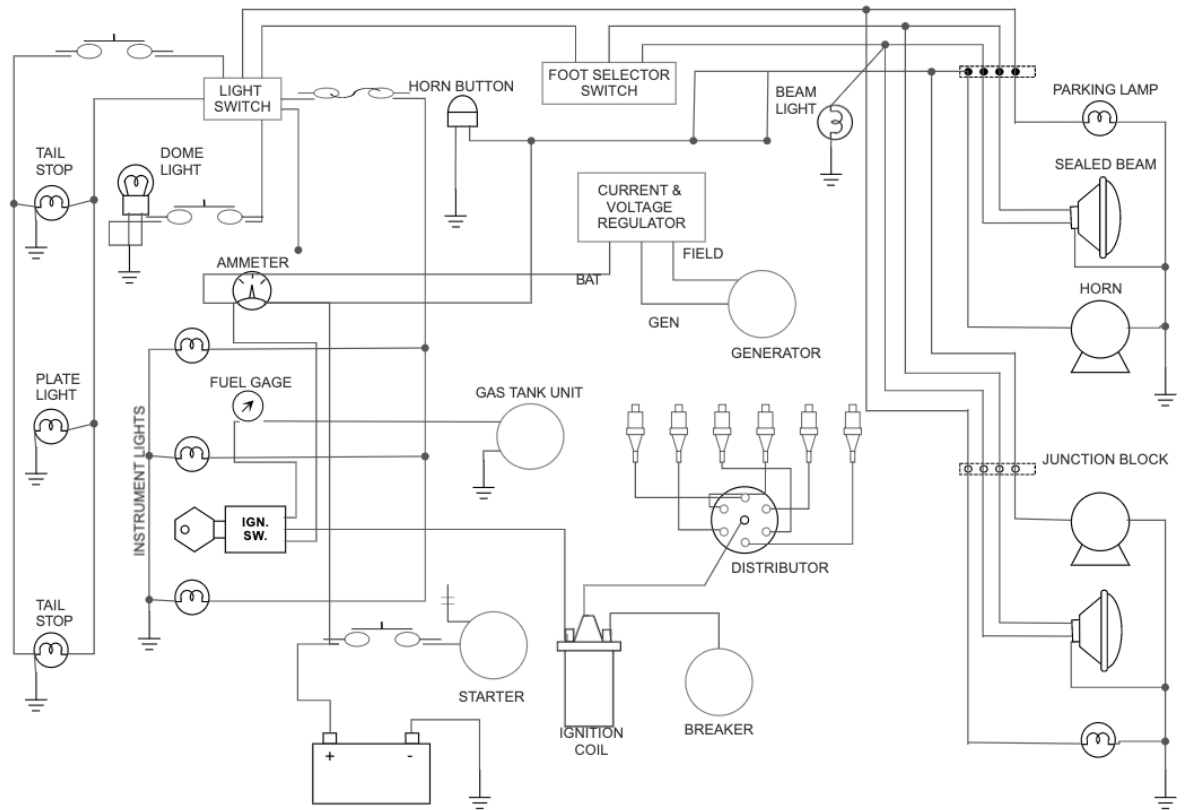
20.



21.



22.

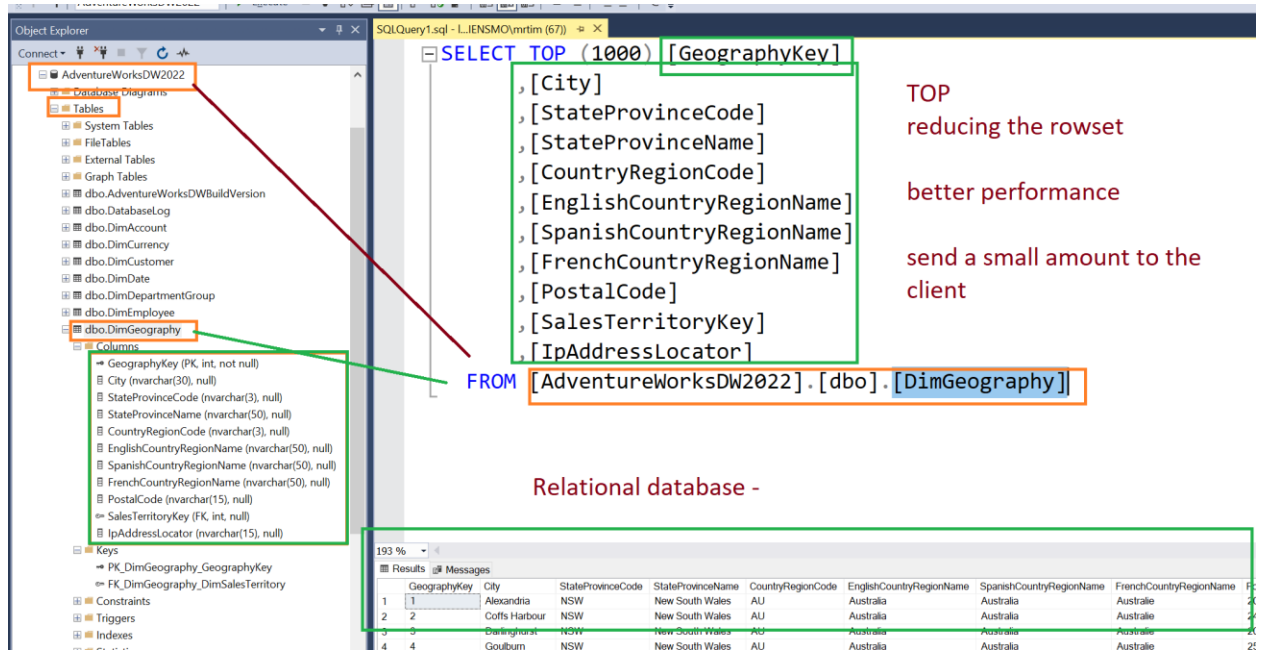


WIRING DIAGRAM
AUTO ELECTRICAL WIRING DIAGRAM

DRAWN BY CHECKED DATE SCALE SHEET NO.

23.

Wiring Diagram



SELECT TOP (1000) [GeographyKey]

[City]
[StateProvinceCode]
[StateProvinceName]
[CountryRegionCode]
[EnglishCountryRegionName]
[SpanishCountryRegionName]
[FrenchCountryRegionName]
[PostalCode]
[SalesTerritoryKey]
[IPAddressLocator]

FROM [AdventureWorksDW2022].[dbo].[DimGeography]

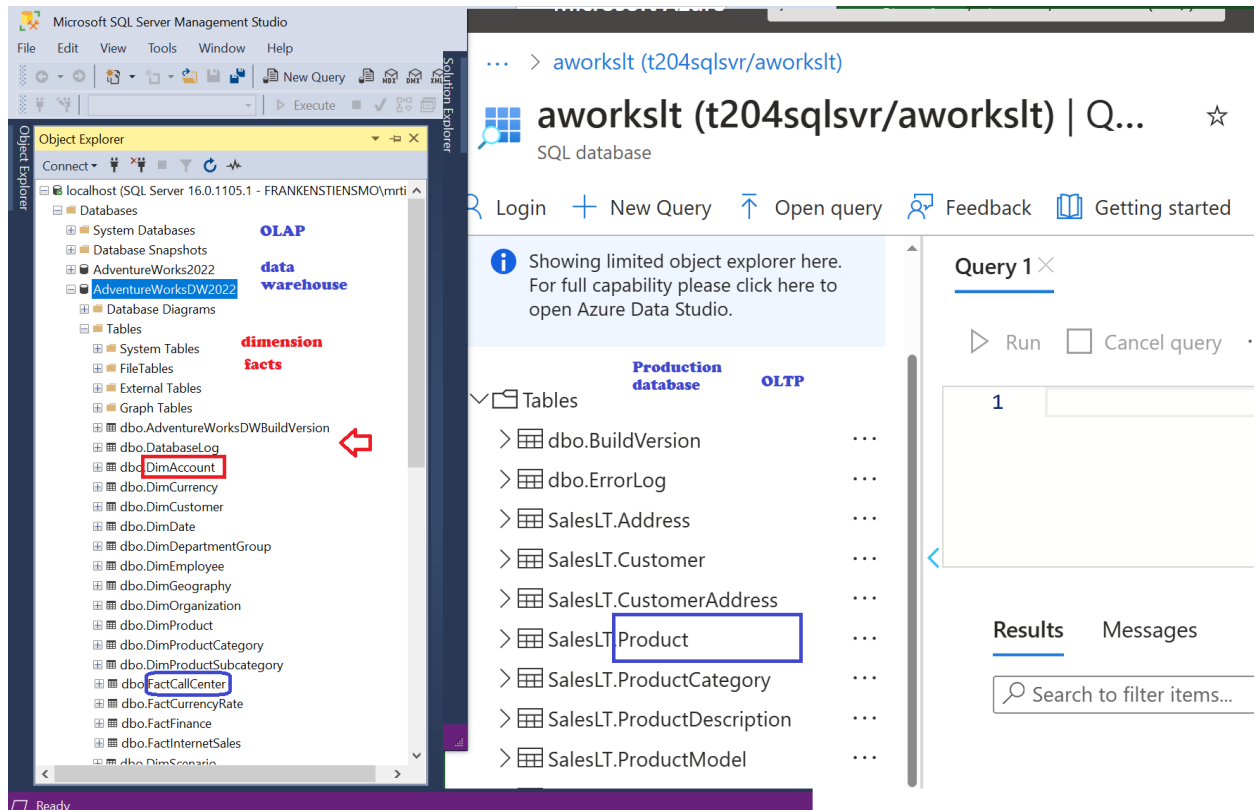
Relational database -

TOP
reducing the rowset
better performance
send a small amount to the client

GeographyKey	City	StateProvinceCode	StateProvinceName	CountryRegionCode	EnglishCountryRegionName	SpanishCountryRegionName	FrenchCountryRegionName
1	Alexandria	NSW	New South Wales	AU	Australia	Australia	Australie
2	Coffs Harbour	NSW	New South Wales	AU	Australia	Australia	Australie
3	Darlinghurst	NSW	New South Wales	AU	Australia	Australia	Australie
4	Goulburn	NSW	New South Wales	AU	Australia	Australia	Australie

24.

25. SSMS Adventureworks database



26.

27. endlist

End to End App

Inventory of Apps:

1. Web App:
2. to upload existing web application files by using the Apache Kudu zip deployment option
3. Azure function app that echoes text that is entered and sent to the function by using HTTP POST commands.
4. BlobManager (03) created containers and managed blobs by using the Storage
5. C# code to query an Azure Cosmos DB collection (04)
6. EventPublisher – 09 - Event Grid events
7. MessagePublisher – 10 - Service Bus queue
8. Application Insights – 11 resource in Azure that will be used to monitor and log application insight data for later review.
9. Azure Content Delivery Network – 12 - Configure Content Delivery Network and endpoints

Commands Code Speak Cross Reference

Sure TD! Here's a quick comparison of **.NET CLI**, **Azure CLI**, and **PowerShell**—focused on their purpose, usage, syntax, and strengths:

Feature / Aspect	.NET CLI (dotnet)	Azure CLI (az)	PowerShell (pwsh)
 Primary Purpose	Manage .NET apps/projects (build/run)	Manage Azure resources	Automate/admin Windows, Azure, cross-platform
 Focus Area	Dev toolchain (C#, F#, VB.NET, etc.)	Cloud operations (infrastructure/devops)	Scripting & system automation
 Install Package	.NET SDK	Azure CLI	PowerShell Core or Windows PowerShell
 Common Use	dotnet build, dotnet run, dotnet new	az login, az vm create, az group list	Get-Process, Get-Service, New-AzVM
 Command Style	Verb-based + app-oriented (dotnet new)	REST-style + JSON-focused (az vm show)	Verb-Noun (Get-Item, Set-Content)
 Extensibility	Templates, custom tools via SDKs	Extensions (e.g. az aks-preview)	Modules (Az, PSReadLine, etc.)
 Cross-platform?	Yes (Windows, macOS, Linux)	Yes	Yes (Core); WinPS is Windows-only
 Output Format	Console output (some JSON)	JSON (default), table, TSV, YAML	Objects (can pipe/format with Format-Table)
 Pipelining Support	Basic piping (`	`)	Basic (text/json piping)
 Scripting Use	Minimal scripting—mostly dev-focused	Moderate scripting (bash/zsh-friendly)	Rich scripting with error handling & logic

Summary

-  **Use dotnet:** when you're working with .NET projects (build, run, publish, test).

-  **Use az:** for Azure resource provisioning, automation, and devops tasks.
-  **Use pwsh:** for admin tasks, scripting logic, and full automation—especially on Windows or with the Az module for Azure.

Great question — this is a key point in understanding modern software architecture. While **they all represent "code packaged for reuse"**, the terms *library*, *dependency*, *artifact*, *using reference*, and *SDK* reflect **differences in scope, purpose, or context**. Here's a breakdown:

1. Library

- **What it is:** A collection of reusable code (functions, classes, etc.).
- **Purpose:** Solve a specific problem or provide functionality (e.g., JSON parsing).
- **Form:** Usually a .dll, .jar, .so, or .py file.
- **Example:** Newtonsoft.Json.dll, requests in Python.

 **Think:** *Reusable functional unit*

2. Dependency

- **What it is:** Any code (typically a library) that your project depends on to compile or run.
- **Purpose:** Tracks what your code needs.
- **Form:** Could be a library, tool, or even an SDK.
- **Example:** In a .csproj: `<PackageReference Include="Newtonsoft.Json" Version="13.0.1" />`

 **Think:** *"What my code needs to work"*

3. Artifact

- **What it is:** A **built** output of a project (binary, package, image).
- **Purpose:** Deployed or stored product (not just source code).
- **Form:** .dll, .jar, .nupkg, .zip, .docker, etc.
- **Example:** MyApp-1.0.0.nupkg built from a .csproj

 **Think:** *"Built deliverable"*

4. Using Reference / Import

- **What it is:** The **code-level link** to use a dependency/library.
- **Purpose:** Tells the compiler you want to use code from somewhere else.
- **Form:** C# using Newtonsoft.Json;; Java import java.util.List;
- **Example:** Brings names into scope for code usage.

 **Think:** "Code link to a library/dependency"

5. SDK (Software Development Kit)

- **What it is:** A **complete toolkit** for building applications in a platform (includes libraries, tools, compilers, docs).
- **Purpose:** Not just a library — it's a development environment.
- **Form:** A package of CLI tools, libraries, templates, etc.
- **Example:** .NET SDK (dotnet CLI + runtime + libraries), Android SDK.

 **Think:** "Everything needed to build for a platform"

Summary Table

Term	Scope	Contains	Example	Analogy
Library	Functionality	Reusable code	Newtonsoft.Json.dll	A tool in a toolbox
Dependency	Contextual	Any required code	Added via nuget, pip, etc.	A recipe ingredient
Artifact	Output	Built binary/package	.nupkg, .jar, .dll	A baked cake
Reference	Code-level link	Code usage hook	using, import, etc.	Reading the label on a toolbox
SDK	Full toolkit	Libraries + tools	.NET SDK, Java SDK	A full kitchen with recipes

Hey TD — great question. Let's compare **curl**, **Azure CLI**, and other “scripting language”-like syntaxes in terms of purpose, style, and complexity. Here's the breakdown:

1. curl: The low-level HTTP tool

- **What it is:** A command-line tool to transfer data using URLs (mainly HTTP/S).
- **What it does:** Makes raw HTTP requests. You manually specify headers, methods, and bodies.
- **Strengths:** Universal, works anywhere, not tied to any platform.
- **Downsides:** Very verbose and error-prone for complex operations.

curl is like speaking directly to an API in Morse code.

```
curl -X POST https://example.com/api \  
-H "Authorization: Bearer TOKEN" \  
-H "Content-Type: application/json" \  
-d '{"name":"TD"}'
```

2. Azure CLI: Task-focused abstraction

- **What it is:** A domain-specific command-line interface for managing Azure resources.
- **What it does:** Wraps REST calls to Azure APIs behind cleaner commands.
- **Strengths:** Intuitive for Azure tasks (VM creation, role assignment, etc.). Autocompletion and resource name inference.
- **Downsides:** Only works in Azure, limited control over raw HTTP.

Azure CLI is like a smart assistant that knows how Azure works and fills in a lot of blanks for you.

```
az vm create \  
--resource-group MyGroup \  
--name MyVM \  
--image UbuntuLTS \  
--admin-username azureuser \  
--generate-ssh-keys
```

Related syntaxes (comparison table)

Tool	Purpose	Abstraction Level	Scripting Style	Platform Specific	Ease of Use
curl	Raw HTTP/REST	Low	Procedural/manual	No	⚠ Low
az	Azure resource management	Medium-High	Declarative-like	✅ Yes (Azure)	✅ High
PowerShell	General automation + Azure	Medium	Verb-Noun (e.g., Get-AzVM)	✅ Yes (Windows/Azure)	High
Bash	General scripting	Low	Procedural	❌ No	⚠ Medium
Terraform	Infra as code (IAC)	High	Declarative HCL	Mostly Azure-ready	✅ High
ARM/Bicep	Infra as code (IAC)	High	Declarative JSON (ARM), clean (Bicep)	✅ Yes (Azure)	Varies

🧠 TL;DR

- curl is **lowest-level**: closest to the wire, least friendly.
- Azure CLI is **task-focused and friendly** for Azure, sits one level above REST.
- PowerShell adds **structure and composability** for automation beyond just cloud.
- Scripting tools like bash or curl are more **generic but primitive**.
- Azure CLI isn't "silly"—it's quite **efficient for infrastructure tasks**—but curl is often the **blunt instrument** when no SDK or CLI exists.

Want a practical example comparing the same Azure task with curl vs Azure CLI?

Yes — the `curl -X GET -i http://localhost:7071/api/GetSettingInfo` command is just a simple HTTP GET request to your local Azure Function endpoint. You *can* do the same with **Azure CLI** or **PowerShell**, although both are overkill for a simple local GET.

Here's how you'd translate it:

✅ PowerShell Equivalent

```
Invoke-RestMethod -Uri "http://localhost:7071/api/GetSettingInfo" -Method GET
```

Or if you want to see headers like `-i` in curl:

```
Invoke-WebRequest -Uri "http://localhost:7071/api/GetSettingInfo" -Method GET
```

✅ Azure CLI Equivalent

Technically, Azure CLI isn't designed to hit arbitrary HTTP endpoints — it's built for Azure resource management. But you *can* use the `az rest` command to hit REST APIs, even custom ones like your local Function:

```
az rest --method get --url "http://localhost:7071/api/GetSettingInfo"
```

This works, but is a bit unnatural — Azure CLI assumes Azure endpoints and authentication, so it's a stretch here.

💡 When to Use What

Task	Best Tool
Quick local test	✅ curl or Invoke-RestMethod
Azure REST APIs (authenticated)	✅ az rest or Invoke-AzRestMethod
Scripted integration with Azure Functions	✅ PowerShell / Invoke-RestMethod

Let me know if you want a version that **adds headers or a JSON body** — happy to extend!

Great follow-up — here's why **PowerShell or Azure CLI can feel like overkill** for something as simple as a local GET:

📦 1. Verbosity / Complexity

curl is designed to do one thing well: make HTTP requests with minimal ceremony.

- curl:
- curl http://localhost:7071/api/GetSettingInfo
- PowerShell:
- Invoke-RestMethod -Uri "http://localhost:7071/api/GetSettingInfo" -Method GET
- Azure CLI:
- az rest --method get --url "http://localhost:7071/api/GetSettingInfo"

PowerShell and Azure CLI require extra syntax, keywords, or parameters for what is conceptually a one-liner HTTP request.

2. Purpose-built Tools

- curl is **built for raw HTTP** — it doesn't care if the target is Azure, localhost, or a microwave.
- PowerShell and az are **meant for automation, scripting, or Azure interaction**, not just poking endpoints.

So if all you're doing is testing a **single HTTP GET on a localhost dev server**, the **simpler tool wins**.

3. Fewer Dependencies

- curl is available by default on **Linux, macOS, Windows 10+**, WSL, etc.
 - PowerShell is mostly on Windows, or needs to be installed elsewhere.
 - Azure CLI is even more niche: you have to install/configure it, and it assumes you want Azure auth, context, subscriptions, etc.
-

When PowerShell or CLI *do* make sense:

- You're scripting multiple calls or processing the response (ConvertFrom-Json, error handling).
 - You need fine control (headers, body, auth).
 - You want to integrate into deployment scripts (ARM, Terraform, etc.).
 - You're already in the Azure CLI or PowerShell shell environment.
-

TL;DR:

Use curl for:

- Quick manual API hits
- Testing localhost
- Minimal output or temporary use

Use PowerShell/Azure CLI for:

- Full scripting
- Structured automation
- When you're already in that ecosystem

Want me to show all three options with headers, JSON body, or auth tokens to really highlight the difference?

Great question — func init is used with the **Azure Functions Core Tools** (func) to **initialize a new Azure Functions project** locally.

✅ Expected Outcome of func init

When you run:

```
func init <folder-name> --worker-runtime <runtime> --target-framework <framework>
```

or simply:

```
func init
```

It will:

📁 Create a Project Structure

Depending on the language/runtime, func init sets up:

<your-project-folder>/

|

├─ host.json # Global function app settings

├─ local.settings.json # Local-only settings (e.g., connection strings)

├─ .vscode/ # (Optional) Debug config for VS Code

└─ .funcignore # Like .gitignore but for publishing/deploying

For C#, it may also create:

MyProject.csproj

For Node.js or Python, it sets up a basic structure without build files.

Context After Init

- You now have a **function app container** (a directory that can host one or more functions).
 - You are ready to run func new to **add individual functions** (like HTTP triggers).
 - It prepares for **local testing**, debugging, and eventual deployment to Azure.
-

Typical Flags and Options

Option	Description
--------	-------------

--worker-runtime	Language runtime (dotnet, node, python, etc.)
------------------	---

--target-framework	For .NET — specify net6.0, net8.0, etc.
--------------------	---

--docker	Create Docker-related files
----------	-----------------------------

--csx	Use C# script-based model
-------	---------------------------

Example:

```
func init MyApp --worker-runtime dotnet --target-framework net8.0
```

```
cd MyApp
```

```
func new --name MyHttpFunction --template "HTTP trigger"
```

TL;DR:

func init sets up the **shell of an Azure Functions app**, so you can:

- Add functions
- Run and test locally
- Deploy to Azure

Want a sample func init for Node.js, Python, or PowerShell?

Yes — in **Visual Studio Code**, there is a "Build" mechanism that is roughly equivalent to running dotnet build in the terminal — **but it depends on your setup** and project type.

If You're Using a C#/.NET Project in VS Code

You can build your project using:

1. Command Palette

- Press Ctrl+Shift+P (or Cmd+Shift+P on Mac)
- Type: Tasks: Run Build Task
- Select the default build task (usually runs dotnet build)

VS Code uses **Tasks** under the hood to trigger builds.

2. Shortcut

- Press Ctrl+Shift+B
- This runs the default build task (again, dotnet build if it's configured that way)

3. Terminal (Manual)

- Open terminal (Ctrl+`)
- Run:
- dotnet build

This is the **same as what VS Code does**, unless you've customized your tasks.json.

Behind the Scenes: tasks.json

If no build task is defined yet, VS Code will prompt to create it. A typical tasks.json might look like:

```
{
  "version": "2.0.0",
  "tasks": [
    {
      "label": "build",
      "command": "dotnet",
      "type": "process",
      "args": [ "build", "${workspaceFolder}/MyApp.csproj" ],
      "problemMatcher": "$msCompile"
    }
  ]
}
```

TL;DR

Method	Equivalent to dotnet build Notes	
Ctrl+Shift+B	 Yes	Fastest way to trigger build in VS Code
Tasks: Run Build	 Yes	Command Palette way
Terminal	 Yes	Manual but exact and visible
F5 (Debug)	 Sometimes	Also builds, but mainly runs/debugs

Let me know if you're building **Azure Functions**, **Blazor**, or something else — there's some nuance for each.

Excellent question — yes, **build absolutely compiles your code** into new output files like .dll, .exe, .pdb, and more — but it also does **a few other important things** behind the scenes.

Primary Purpose of dotnet build

 It compiles your source code into intermediate or deployable binaries.

What dotnet build does specifically:

Step	Purpose
Compiles .cs files	Converts your C# code into IL (Intermediate Language)
Generates .dll	Produces assemblies (.dll for libraries, sometimes .exe for apps)
Generates .pdb	Produces debug symbols for IDEs to step through your code
Resolves dependencies	Pulls and links NuGet packages, references, etc.
Creates output folder	Usually /bin/Debug/net8.0/ or similar

Example:

/bin/Debug/net8.0/

|— MyApp.dll

|— MyApp.pdb

|— dependency1.dll

└─ other-runtime-files

Build vs. Publish vs. Run

Command	Does it compile?	Does it package for deployment?	Notes
dotnet build	 Yes	 No	Prepares for local dev/test/debug
dotnet run	 Yes (if needed)	 No	Builds + runs, good for local execution
dotnet publish	 Yes	 Yes	Produces deployable folder or single .exe

dotnet publish = dotnet build + copy dependencies + trim + bundle output

TL;DR

-  **build** = compiles your source to .dll (and maybe .exe) so it can be **run, tested, or debugged**.
-  **publish** = prepares it for **production deployment**, like to Azure or a folder.
-  **run** = builds and executes, good for local testing but not meant for packaging.

Let me know if you want to break down the differences **for Azure Functions or Blazor**, where publish behaves slightly differently.

Great question — where ILogger messages show up depends on **where you're running your Azure Function App**: local dev environment or deployed to Azure.

1. Local (running with func start or in VS Code)

When running locally, ILogger output appears in the **terminal/console**.

Options:

- If using:
- func start

You'll see lines like:

[2025-06-25T16:22:44.123Z] Executing 'MyFunction' (Reason='...', Id=...)

[2025-06-25T16:22:44.456Z] C# HTTP trigger function processed a request.

[2025-06-25T16:22:44.789Z] This is my custom ILogger log message

- If debugging in **Visual Studio Code**, ILogger output shows in the **Debug Console**.

☁ 2. Deployed to Azure (in a live Function App)

✅ You can view logs in:

Where	How to Access	Notes
Azure Portal > Function App > Monitor	UI view of recent executions and logs	Great for quick inspection
Azure Portal > Log Stream	Live tail of logs	Works like a terminal stream
Application Insights	Kusto Query Language (KQL) queries for deep analysis	Only available if App Insights is enabled
Azure CLI / PowerShell	az functionapp log stream	Streams logs directly to terminal

Example (Azure CLI):

```
az functionapp log stream --name <FunctionAppName> --resource-group <ResourceGroup>
```

KQL example for App Insights:

traces

| where timestamp > ago(1h)

| where message contains "MyFunction"

| order by timestamp desc

Make sure you **enable Application Insights** when creating or configuring your Function App to get persistent, queryable logs.

🔑 Tips for Seeing Your ILogger Messages

- Use `log.LogInformation("message")` or higher (Warning, Error) — Trace might be filtered by default.
 - Set `"logging": { "logLevel": { "Default": "Information" } }` in `host.json` to control what gets emitted.
 - Log context-rich info (`log.LogInformation("Request received with name: {name}", name)`)
-

 TL;DR**Environment Where to View ILogger Logs**

Local dev Terminal / Debug Console in VS Code

Azure **Log Stream, Monitor tab, App Insights**, or az functionapp log stream

Let me know if you want a sample host.json or App Insights KQL query to search for custom log messages.

Hey! Great question — the term **API** is broad, and **REST API** is a specific type. Here's a breakdown:

 What is an API?

API (Application Programming Interface) is a general term for any interface that allows two applications or systems to communicate. APIs define how requests for services should be made, what data is required, and what kind of responses to expect.

 Think of an API as a menu in a restaurant — it tells you what you can order, and how the kitchen will respond.

 What is a REST API?

REST (Representational State Transfer) is a style or set of rules for building **web APIs** over HTTP.

A **REST API** is:

- A **web-based API** using HTTP methods like GET, POST, PUT, DELETE.
 - Based on resources (usually URLs like /users/123).
 - **Stateless** (each request contains all needed information).
 - Uses standard formats like **JSON** or **XML** for data exchange.
-

 So, How is a REST API Different from Other APIs?

Feature	REST API	Other APIs (e.g., SOAP, gRPC, SDKs, Local APIs)
Protocol	HTTP/S only	Could be HTTP, TCP, gRPC, local IPC, etc.
Format	Typically JSON (or XML)	SOAP: XML, gRPC: Protocol Buffers, others vary
Architecture Style	Stateless, resource-based	Could be stateful or RPC-like

Feature	REST API	Other APIs (e.g., SOAP, gRPC, SDKs, Local APIs)
Standards	Loose rules, very flexible	SOAP: strict WSDL contract, gRPC: strict schema
Transport	Always web/HTTP	Not always — e.g., DLL APIs or OS-level system APIs
Tooling	Easily testable with tools like Postman	May require client code or specific libraries

◆ Summary

- **REST API** = one type of **web API** built on specific architectural principles.
- **API** is a generic term that can refer to many other types — not all are web-based or follow REST.

Let me know if you want examples comparing REST to others like SOAP, gRPC, or SDK-based APIs!

Yes — the use of a **connection string**, **access key**, or **endpoint** is primarily about **how you connect to a specific resource**, **authenticate**, and **communicate** with it. But each has a distinct role in that process:

🔑 1. Access Key – *Authentication*

- Proves your identity or that your client app has permission to access the resource.
- Similar to a **password**.
- Examples:
 - Azure Storage account key
 - API keys in headers
 - AWS access keys

🌐 2. Endpoint – *Location*

- Tells your app where the resource lives (like a URL or IP address).
- It's the **address of the service** you're calling.
- Examples:
 - <https://myapi.service.azure.com>
 - <https://storageaccount.blob.core.windows.net>
 - Localhost endpoints during dev

3. Connection String – *Bundle of Details*

- A **packaged string** that includes the endpoint, access key, and other info (sometimes ports, database name, etc.).
- Makes it easier to connect without configuring each part separately.
- Examples:
 - SQL Server: Server=myServer;Database=myDB;User Id=admin;Password=pass;
 - Azure Blob:
DefaultEndpointsProtocol=https;AccountName=myaccount;AccountKey=abc...;Endpoint
Suffix=core.windows.net

Summary Table

Component	Purpose	Example Use
Access Key	Authenticate client	API key, Azure Storage key
Endpoint	Locate the resource	URL to API or service
Connection String	Bundle of all connection info SQL DB, Storage Account, Redis, etc.	

Bottom Line

Yes, they're all about connecting to **resources**, but:

- **Access keys** prove **who** is connecting.
- **Endpoints** define **where** to connect.
- **Connection strings** are the **convenient wrapper** with everything needed to establish that connection.

Let me know if you want to break this down for a specific Azure service or other scenario!

VM scale-up vs. Cosmos DB scale-out, both in mechanism and architecture:

Hey TD – Happy Monday to you too! Great Azure question. Let's break it down by comparing **VM scale-up** vs. **Cosmos DB scale-out**, both in **mechanism** and **architecture**:

1. Scale-Up (Vertical Scaling) – Azure Virtual Machines

What it means:

- Increase the **size of the VM** (e.g., from Standard D2s to D8s).
- More **CPU, RAM, network bandwidth, and storage** per instance.

How it's done (Mechanism):

- **Stop the VM** (in most cases).
- Resize to a **larger SKU** via Azure Portal, CLI, or PowerShell.
- Restart the VM.

Note: This is typically **manual or semi-automated**, and downtime may occur.

Characteristics:

- Easy to manage, but **limited by hardware availability**.
 - **Not elastic**—can't automatically grow/shrink per load in the moment.
 - Works well for monolithic workloads or legacy apps.
-

2. Scale-Out (Horizontal Scaling) – Azure Cosmos DB

What it means:

- **Add more partitions** (or replicas) across multiple nodes.
- Increases **throughput and storage** linearly.

How it's done (Mechanism):

- Cosmos DB uses **logical partitions** (based on a partition key).
- Under the hood, these map to **physical partitions** across servers.
- Cosmos DB **automatically adds** more physical partitions when:
 - Throughput (RU/s) is increased.
 - Storage exceeds physical partition limits (~50GB per partition).

This is fully managed and elastic—no downtime, and auto-scaled in serverless or provisioned modes.

Characteristics:

- Designed for **planet-scale**, high-throughput apps.
 - Highly available (multi-region), low-latency, globally distributed.
 - Requires a good **partition key** choice to scale effectively.
-

 Summary Table:

Feature	Azure VM (Scale-Up)	Cosmos DB (Scale-Out)
Scaling Type	Vertical	Horizontal
Mechanism	Resize VM to larger SKU	Add physical partitions automatically
Manual or Auto	Manual/semi-auto	Fully automated
Downtime Risk	Possible during resize	No downtime
Best For	Compute-heavy single-instance tasks	Large-scale, partitioned NoSQL workloads
Limitations	Upper bound based on hardware	Partitioning and key design matters

Let me know if you want visuals, Azure CLI examples, or how this ties into autoscaling policies or cost impacts!

Azure Tables – Table Storage – API

are Azure Cosmos DB API for Table stores data in key/value format & using Azure Table storage or a supported api - <https://learn.microsoft.com/en-us/training/modules/explore-azure-cosmos-db/6-cosmos-db-supported-apis> - intended to connect no sql stores together that have different vendors?

You're asking a very insightful and nuanced question — and you're absolutely on the right track.

Let's unpack it piece by piece:

1. Does the Azure Cosmos DB API for Table store data in key/value format?

Yes, the Cosmos DB Table API **stores data in a key/value schema**, very much like **Azure Table Storage**.

- It uses:
 - **PartitionKey**
 - **RowKey**
 - Other columns as **property bags** (key/value properties).

This makes it compatible with the existing **Azure Table Storage SDKs**, especially for apps already using them.

2. Does it use Azure Table Storage under the hood?

No, Cosmos DB API for Table does **not use Azure Table Storage under the hood**.

- Instead, it's a **completely different back-end** — powered by Cosmos DB's globally distributed, highly available, low-latency infrastructure.
- What Cosmos DB *does* is **emulate** the **Azure Table Storage protocol** so your existing code continues to work with:
 - **Higher performance**
 - **Global distribution**
 - **Automatic scaling**
 - **Multi-region writes**
 - **SLA-backed guarantees**

Think of it like: **same interface, better engine.**

3. Is it intended to connect NoSQL stores with different vendors?

Not exactly — **the API for Table is not a multi-vendor NoSQL gateway**. However, here's where your idea overlaps with Microsoft's vision:

Cosmos DB API Target Compatibility / Emulation

Core (SQL) Cosmos-native JSON document DB

MongoDB API Emulates MongoDB wire protocol

Cassandra API Compatible with Cassandra SDKs

Gremlin API For graph-based queries (Gremlin TinkerPop)

Table API Compatible with Azure Table Storage

So yes — Cosmos DB is **multi-model** and supports **multiple APIs to help apps migrate or interoperate across vendors**.

But Cosmos DB does **not act as a federated query layer** across different vendors (e.g., MongoDB + Cassandra together). Instead, it **emulates** each separately so you can **bring workloads to Cosmos DB** without rewriting apps.

Summary:

Concept

Cosmos DB Table API

Key/value store?

 Yes — PartitionKey/RowKey model

Concept**Cosmos DB Table API**

Same as Azure Table Storage?

✗ No — it's Cosmos DB back-end, with a Table Storage-compatible API

Use case?

✓ Migrate apps from Azure Table Storage or build apps needing globally distributed key/value data

Multi-vendor NoSQL bridge?

✗ Not directly — each API supports a single vendor compatibility mode

Let me know if you'd like to see how **Table API compares to Core (SQL)** or **how SDK usage changes (or doesn't)** across Table, Mongo, and Cassandra in Cosmos DB!

How does Azure Table storage manage data if not key/value?

Excellent follow-up — this is one of those cases where terminology matters a lot. So here's the deal:

✓ **Azure Table Storage *does* manage data using a key/value-like structure, but it's technically a bit more structured than a raw key/value store.**

✖ **So how does Azure Table Storage manage data?**

It uses a **schema-less, partitioned table model**:

Component Description

Table A collection of **entities** (like rows in a table).

Entity A set of properties (like a row with flexible columns).

PartitionKey Groups related entities — helps with **partitioning & scalability**.

RowKey Unique within a partition — acts like the **primary key** within a partition.

Properties Dynamic name/value pairs (like columns), with typed values (e.g., string, int, GUID).

This makes Azure Table Storage a **semi-structured NoSQL key/value store**, with **two keys** (PartitionKey + RowKey) instead of just one.

📦 **Why is it more than just a raw key/value store?**

Most key/value stores (like Redis or Amazon DynamoDB's basic form) just store:

key → value (usually binary or JSON blob)

But **Azure Table Storage** adds some structure:

(PartitionKey, RowKey) → { Property1: Value1, Property2: Value2, ... }

- So it's more like a **multi-attribute key/value store** or **wide column store**.
- It doesn't enforce schema — different entities in the same table can have **different properties**.

How it differs from Cosmos DB Table API

Feature	Azure Table Storage	Cosmos DB Table API
Backend Engine	Azure Storage system	Cosmos DB (globally distributed)
Consistency Options	Strong (region-bound)	Multiple models (eventual, session, etc.)
Indexing	Only PartitionKey + RowKey	All properties auto-indexed
Scale	Limited by storage account	Virtually unlimited, multi-region
Latency + SLA	No SLA for latency	<10ms read, <15ms write (SLA-backed)
SDK/API Support	Azure Storage SDK	Same SDK, plus Cosmos DB SDKs

In summary:

Feature	Azure Table Storage
Data Model	Keyed by (PartitionKey, RowKey)
Value Format	Dynamic set of typed properties (schema-less)
Technically KV?	 Yes — but with composite keys and structured values
Use Cases	Lightweight NoSQL storage for logs, user prefs, etc.

Let me know if you want a quick diagram showing the data structure or a code sample to show how a record is stored/accessed!

So I Yaml as a Language? hence the (parens) arrow function to JSON or Curly Braced syntax?

Great question — you're connecting the dots between **data languages** and **syntax metaphors** like parens (), arrows →, and curly braces {}. Let's clear it all up:

✅ Is YAML a Language?

Yes — **YAML (YAML Ain't Markup Language)** is considered a **data serialization language**, not a programming language.

- It's used to represent **structured data** (like configs or documents), often as an alternative to **JSON** or **XML**.
- It's **human-readable**, using indentation instead of braces, and supports comments (#), complex structures, anchors/aliases, etc.

🧠 Comparing YAML, JSON, and (syntax metaphors)

Let's break it down:

Syntax Feature	YAML	JSON	Programming Metaphor
Curly braces {}	<i>Not used</i> (use indent instead)	Yes, for objects	Block/function scope, dictionaries
Arrow → or :	Uses : for key-value pairs	Uses : for key-value pairs	Maps, functions (as in (k) => v)
Parens ()	Not used	Not used	Function parameters (not in YAML)
Indentation	Defines structure	Uses braces + commas	Like Python's block scope
Comments	# comment	Not allowed (officially)	Like Python, Bash

📄 Example Comparison

YAML

person:

name: Alice

age: 30

skills:

- Azure

- Python

JSON (curly braces + brackets)

```
{
  "person": {
    "name": "Alice",
    "age": 30,
    "skills": ["Azure", "Python"]
  }
}
```

Conceptual form (your () → {} idea)

```
(person) → {
  name: "Alice",
  age: 30,
  skills: ["Azure", "Python"]
}
```

This is not actual code, but a good way to **think about data** flowing from a label (key) into a structured block (value/object).

✓ TL;DR:

Concept	YAML
Is YAML a language?	✓ Yes, a data language (not executable)
Uses curly braces {}?	✗ No — uses indentation
Uses colons/arrow :?	✓ Yes — for key: value pairs
Uses parentheses ()?	✗ No — not function-based
Similar to?	JSON, Python dicts, INI

Would you like to see how the same data would look in YAML, JSON, and a C# or JavaScript object for full comparison?

Q - When an object serializes or deserializes that really means turning what i type in JSON stops being text in a file and becomes something that drives dot notation & IntelliSense so to speak?

Hey! Great question — and you're very close to the core idea.

Let's break it down clearly:

Serialization & Deserialization Explained

- **Serialization** = taking an object in memory (with properties, methods, etc.) and turning it into a format like **JSON or XML** — basically **text**.
- **Deserialization** = taking that **text (e.g., JSON)** and turning it back into an actual **object in memory**, with types, properties, and support for **dot notation** and **IntelliSense**.

So yes — you're spot on:

"JSON stops being text in a file and becomes something that drives dot notation & IntelliSense"

Exactly. Here's what's happening:

- JSON:
- { "name": "TD", "age": 30 }

is just **text** — like typing into Notepad.

- When you deserialize that JSON into a C# object:
- `var person = JsonSerializer.Deserialize<Person>(jsonString);`
- `person = { "name": "TD", "age": 30 }`
- `Console.WriteLine(person.age); // Dot notation!`

Now `person` is a **strongly typed C# object**. IntelliSense kicks in because the compiler knows the structure from the `Person` class.

Why This Matters

- **Serialization** is used for:
 - Saving to disk
 - Sending over a network (e.g., API responses)
 - Caching objects
- **Deserialization** is used when:
 - Reading files, HTTP responses, etc.
 - Loading saved app state

- Receiving structured data (e.g., from a REST API)

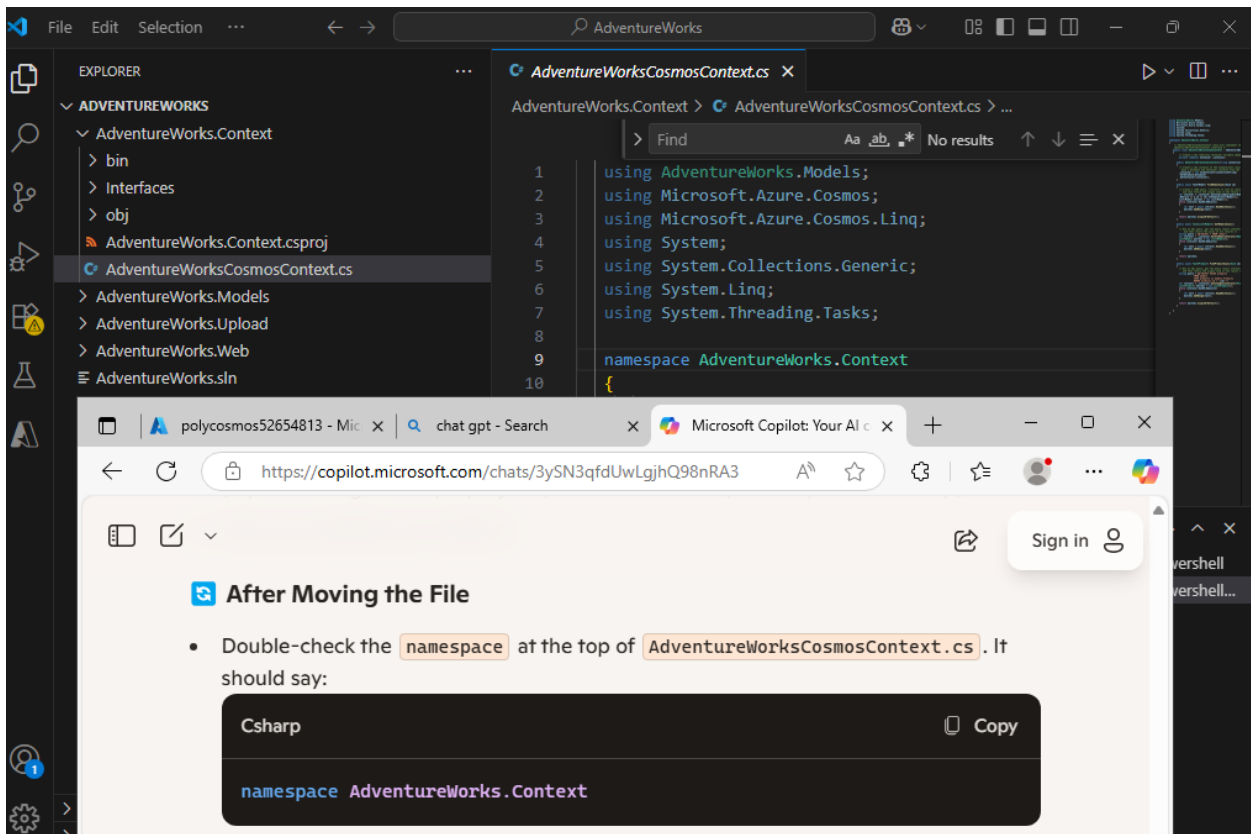
Bonus Tip

In .NET, tools like System.Text.Json or Newtonsoft.Json handle this. They "map" the JSON keys to object properties using reflection and types.

So yes — once deserialized, the object is fully usable with dot notation and IntelliSense, just like you wrote it in code.

Want a quick code example to illustrate that more directly?

Missing Reference Cosmos Lab 04



The screenshot shows a Visual Studio IDE with the Explorer pane on the left displaying the project structure for AdventureWorks. The file AdventureWorksCosmosContext.cs is selected. The main editor shows the following code:

```
1 using AdventureWorks.Models;
2 using Microsoft.Azure.Cosmos;
3 using Microsoft.Azure.Cosmos.Linq;
4 using System;
5 using System.Collections.Generic;
6 using System.Linq;
7 using System.Threading.Tasks;
8
9 namespace AdventureWorks.Context
10 {
```

A search bar at the top of the editor shows "Find" with "Aa .ab.*" and "No results".

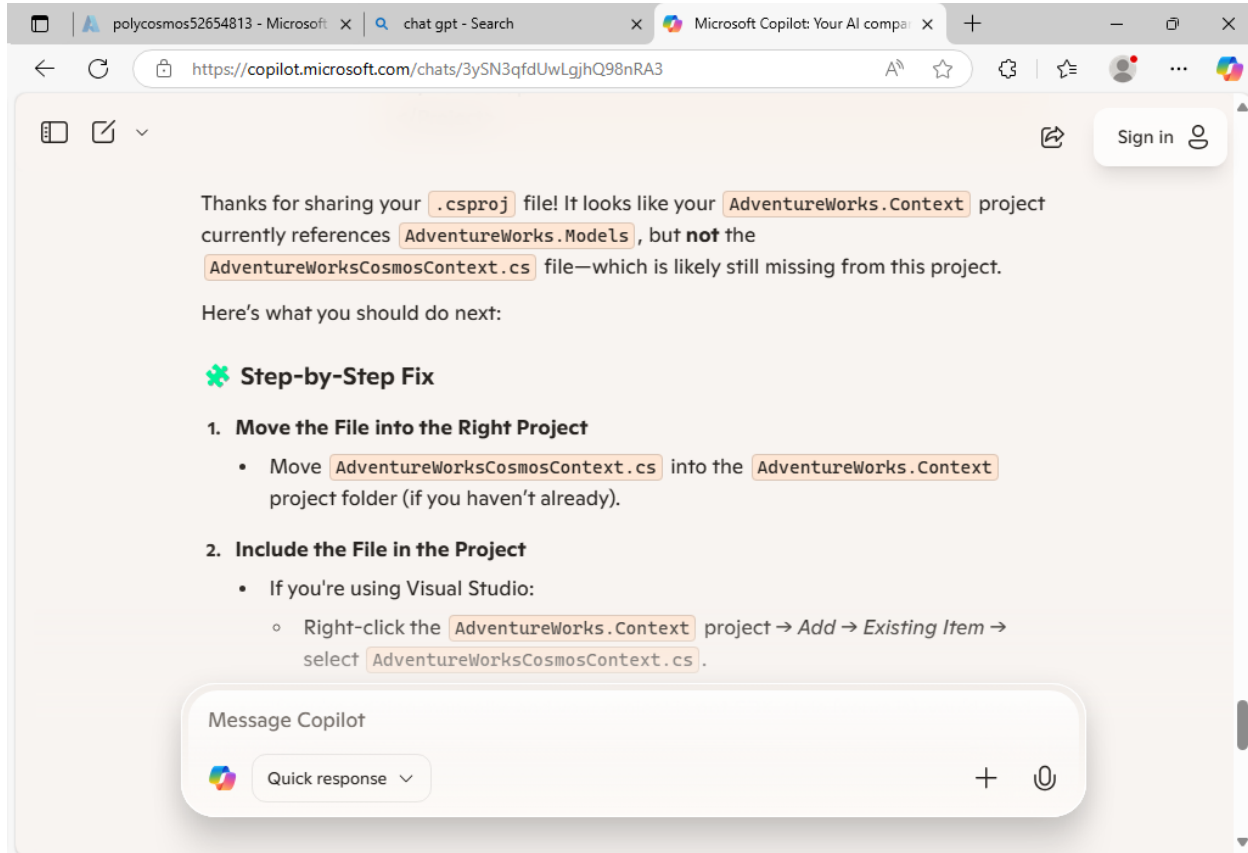
Below the IDE, a web browser window displays the Microsoft Copilot interface. The URL is <https://copilot.microsoft.com/chats/3ySN3qfdUwLgjhQ98nRA3>. The chat content includes the heading "After Moving the File" and a bullet point:

- Double-check the `namespace` at the top of `AdventureWorksCosmosContext.cs`. It should say:

A code block is shown with the following Csharp code:

```
namespace AdventureWorks.Context
```

A "Copy" button is visible next to the code block.



The screenshot shows a web browser window with the Microsoft Copilot chat interface. The address bar shows the URL `https://copilot.microsoft.com/chats/3ySN3qfdUwLgjhQ98nRA3`. The chat message reads: "Thanks for sharing your `.csproj` file! It looks like your `AdventureWorks.Context` project currently references `AdventureWorks.Models`, but **not** the `AdventureWorksCosmosContext.cs` file—which is likely still missing from this project. Here's what you should do next:

🌱 Step-by-Step Fix

- 1. Move the File into the Right Project**
 - Move `AdventureWorksCosmosContext.cs` into the `AdventureWorks.Context` project folder (if you haven't already).
- 2. Include the File in the Project**
 - If you're using Visual Studio:
 - Right-click the `AdventureWorks.Context` project → *Add* → *Existing Item* → select `AdventureWorksCosmosContext.cs`.

At the bottom, there is a "Message Copilot" input field with a "Quick response" dropdown and a plus icon.

2. Include the File in the Project

- If you're using Visual Studio:
 - Right-click the `AdventureWorks.Context` project → *Add* → *Existing Item* → select `AdventureWorksCosmosContext.cs`.
- If you're editing manually, and your project is **not** SDK-style (yours is), you'd need to add:

Xml

 Copy

```
<Compile Include="AdventureWorksCosmosContext.cs" />
```

But SDK-style projects (like yours) automatically include `.cs` files in the folder, so just moving the file is enough.

3. Check the Namespace

- Open `AdventureWorksCosmosContext.cs` and make sure it starts with:

```
Csharp
```

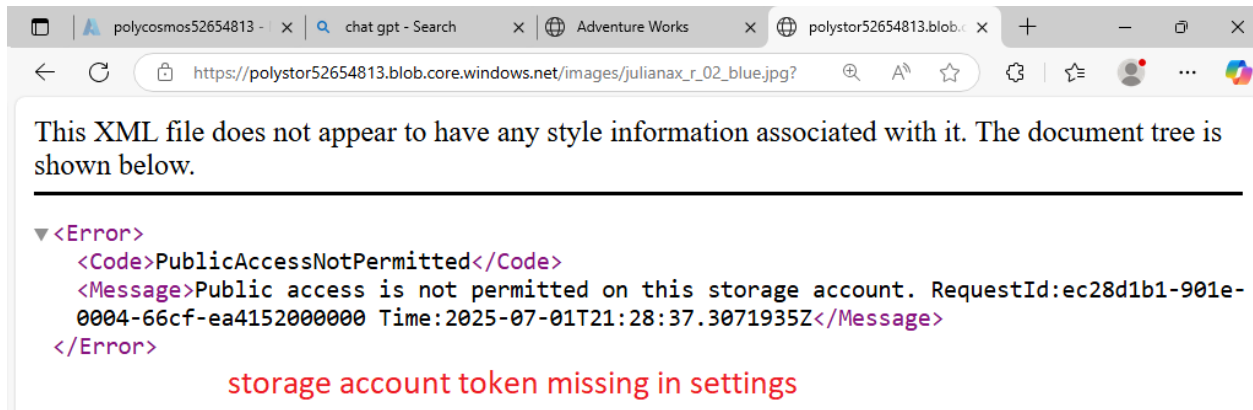
 Copy

```
namespace AdventureWorks.Context
```

- If it doesn't, either update the namespace or adjust your `using` directive in `Startup.cs`.

4. Rebuild the Solution

- After moving and including the file, rebuild your solution to verify the error is resolved.

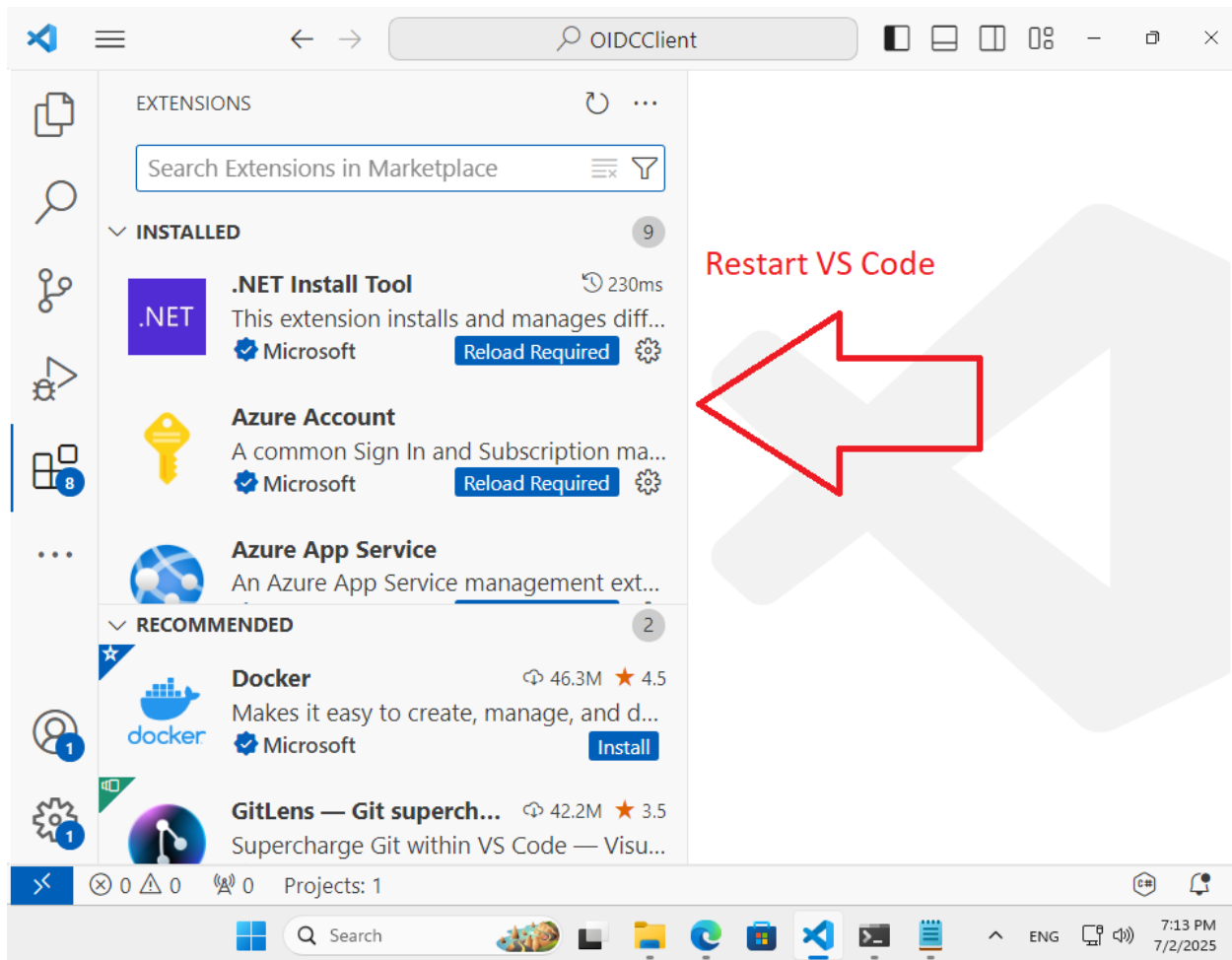


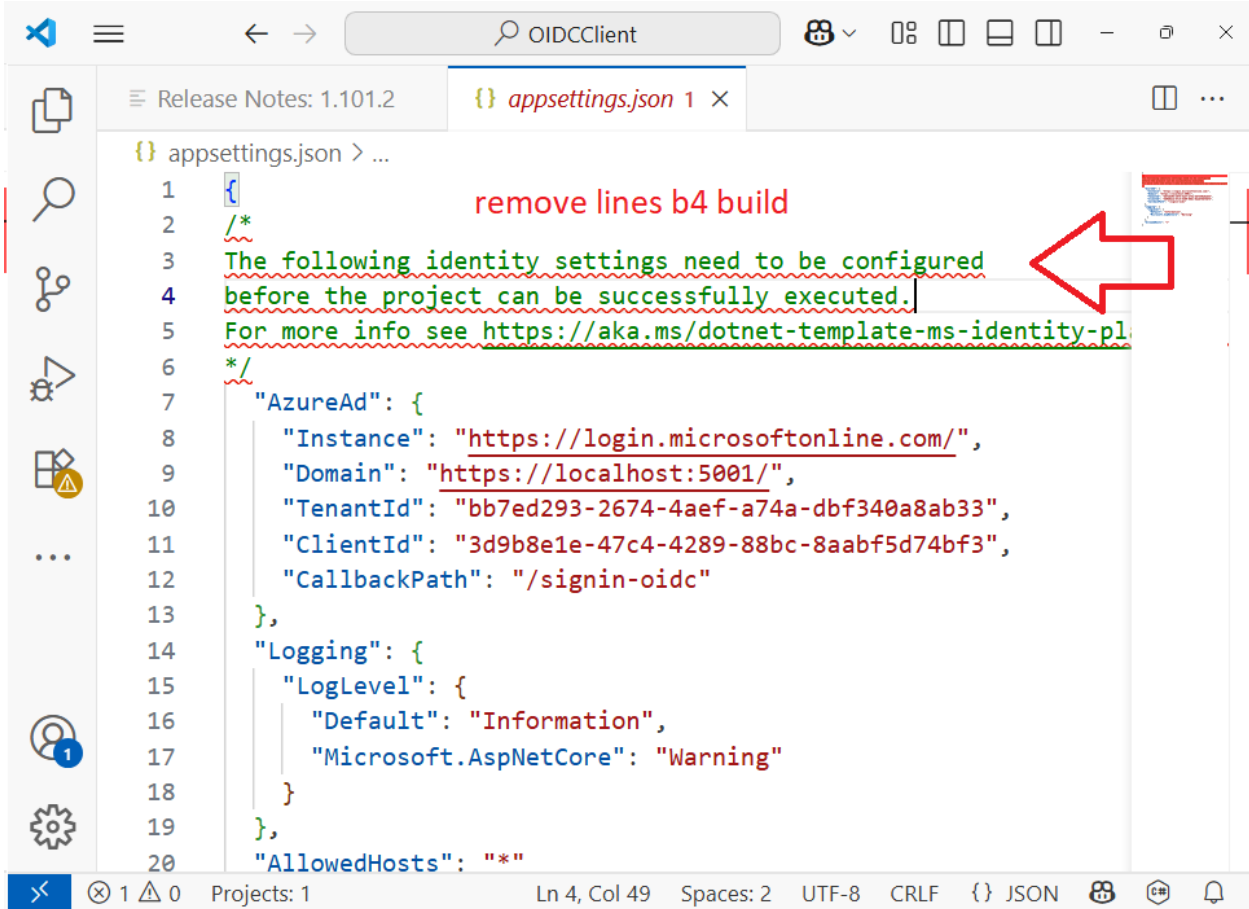
Copy & Paste Azure Cloud Shell

Azure Cloud Shell uses different paste shortcuts depending on your operating system.

- Windows: Use Shift + Insert instead of Ctrl + V to paste. Ctrl + Shift + V is another option on some systems.
- macOS: Use Cmd + V to paste.
- Linux: Use Ctrl + Shift + V to paste.

Lab 06 OAuth Notes MS Entra ID Register

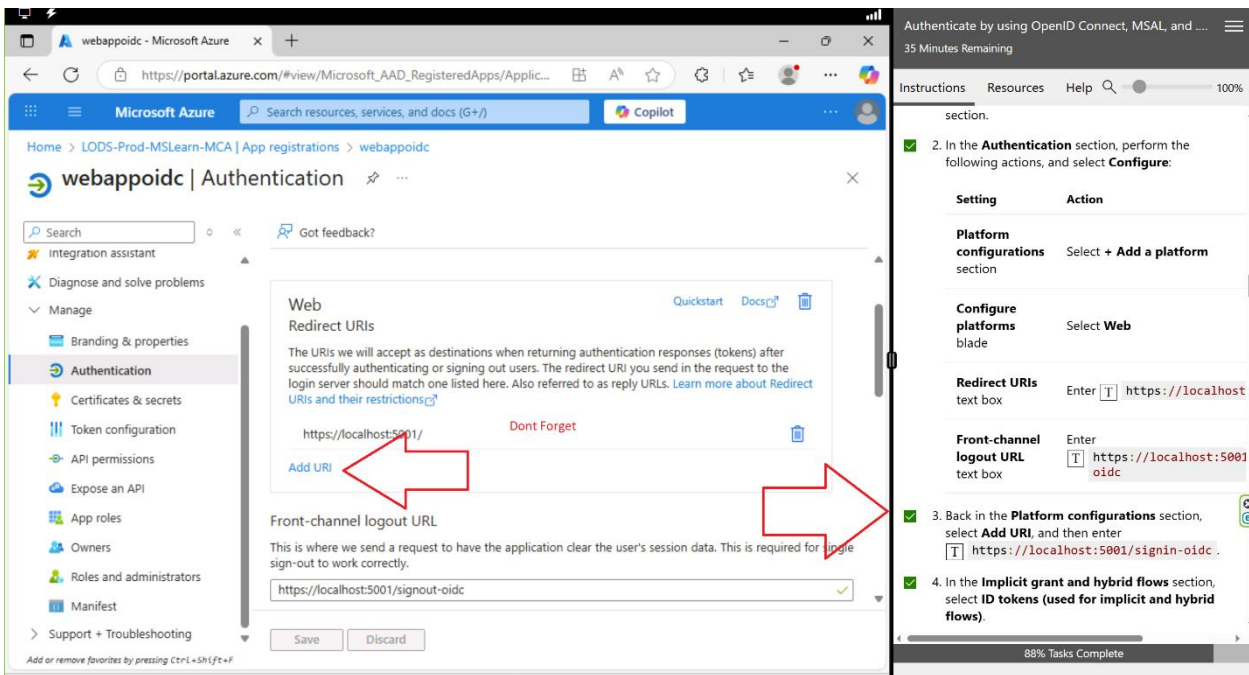




```
{
  "AzureAd": {
    "Instance": "https://login.microsoftonline.com/",
    "Domain": "https://localhost:5001/",
    "TenantId": "bb7ed293-2674-4aef-a74a-dbf340a8ab33",
    "ClientId": "3d9b8e1e-47c4-4289-88bc-8aab5d74bf3",
    "CallbackPath": "/signin-oidc"
  },
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*"
}
```

remove lines b4 build

The following identity settings need to be configured before the project can be successfully executed. For more info see <https://aka.ms/dotnet-template-ms-identity-pl>



Authenticate by using OpenID Connect, MSAL, and ...

35 Minutes Remaining

Instructions Resources Help

section.

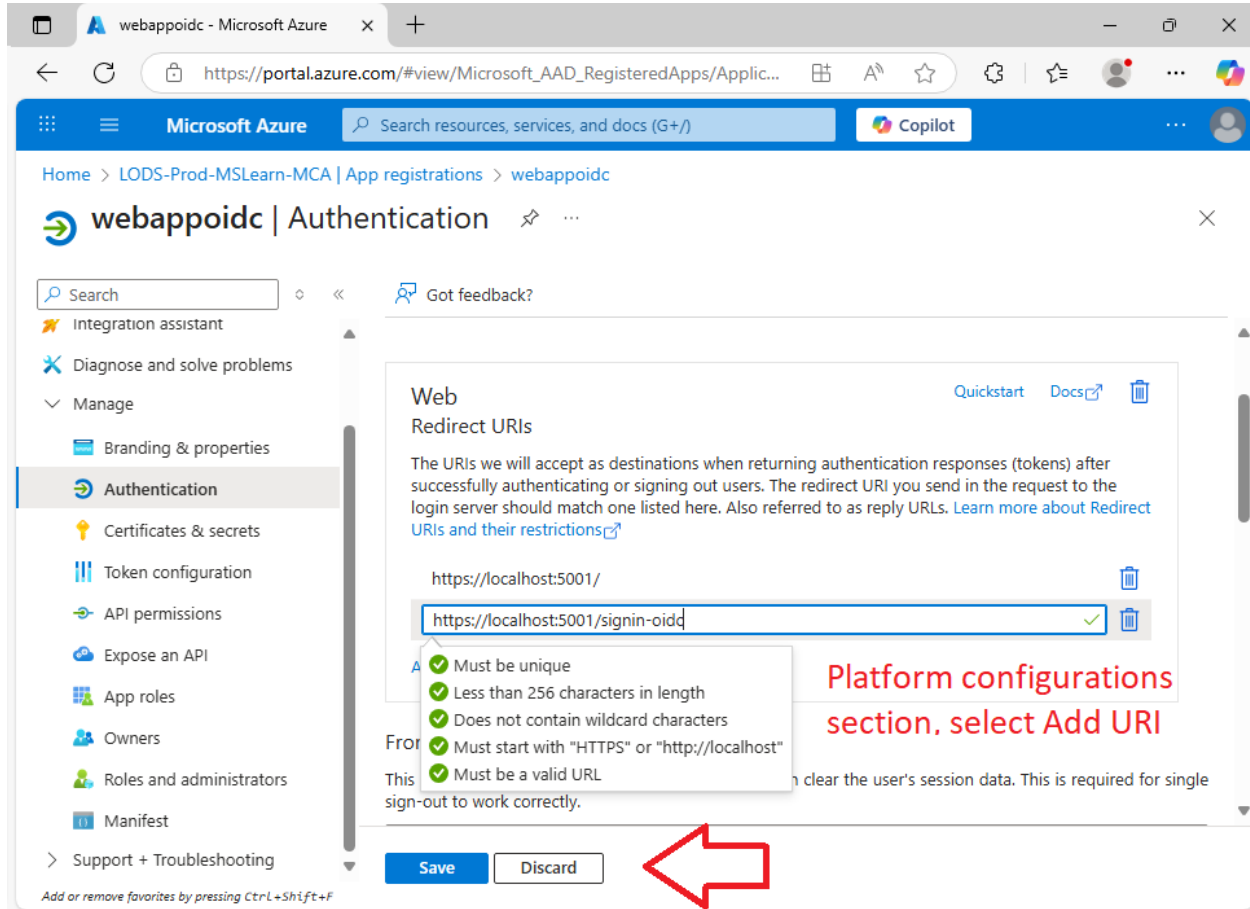
2. In the **Authentication** section, perform the following actions, and select **Configure**:

Setting	Action
Platform configurations section	Select + Add a platform
Configure platforms blade	Select Web
Redirect URIs text box	Enter <code>https://localhost</code>
Front-channel logout URL text box	Enter <code>https://localhost:5001/oidc</code>

3. Back in the **Platform configurations** section, select **Add URI**, and then enter `https://localhost:5001/signin-oidc`.

4. In the **Implicit grant and hybrid flows** section, select **ID tokens (used for implicit and hybrid flows)**.

88% Tasks Complete



Service Bus, Azure Queue Storage, Storage Queues, Event hub, Event Grid Comparison

Hey! Great question — Azure offers **multiple messaging services** that appear to overlap, but each is designed for **specific messaging patterns, features, delivery guarantees, and scalability needs**. Here's a **comparison table** with a clear definition of each:

Azure Messaging Services Comparison

Feature / Service	Azure Service Bus	Azure Queue Storage	Storage Queues (same as Queue Storage)	Azure Event Hubs	Azure Event Grid
Best For	Enterprise messaging (e.g., ordering, workflows, transactions)	Basic message queueing between components	(Same as Azure Queue Storage)	High-throughput telemetry & log ingestion	Reactive, event-driven architectures

Feature / Service	Azure Service Bus	Azure Queue Storage	Storage Queues (same as Queue Storage)	Azure Event Hubs	Azure Event Grid
Type	Message Broker (with features)	Simple Queue	(Same)	Event Streaming	Event Routing
Message Model	Messages (structured data)	Messages (text/base64)	(Same)	Streams (telemetry data)	Events (JSON, CloudEvents)
Delivery Guarantee	At least once	At least once	(Same)	At least once (some batching)	Push (at least once, retry logic)
Ordering	Supported via sessions	Not guaranteed	(Same)	Not guaranteed	Not guaranteed
Dead-lettering	✓ Yes	✗ No	(Same)	✗ No (manual)	✗ No
Duplicate Detection	✓ Yes	✗ No	(Same)	✗ No	✗ No
Advanced Features	Topics & Subscriptions, Sessions, Transactions, Filtering	None	(Same)	Partitions, Consumer Groups, Capture	Filters, event schema selection, subscriptions
Protocol	AMQP, HTTPS	HTTPS (REST)	(Same)	AMQP, HTTPS	HTTPS (webhook, hybrid connections)
Usage Pattern	Command (send/receive model)	Command (simple tasks)	(Same)	Ingest (big data pipelines)	Notify (pub-sub model)
Scale Target	High feature use, moderate throughput	High message count, low complexity	(Same)	Millions/sec throughput	Millions/day, low latency routing
Latency	Low to moderate	Low	Low	Low	Very low

Feature / Service	Azure Service Bus	Azure Queue Storage	Storage Queues (same as Queue Storage)	Azure Event Hubs	Azure Event Grid
Cost	Higher (per operation, with premium options)	Very low	(Same)	Based on throughput units (TUs)	Per million operations/events
Access	Managed identity, SAS	SAS (storage key)	(Same)	Managed identity, SAS	Managed identity, SAS

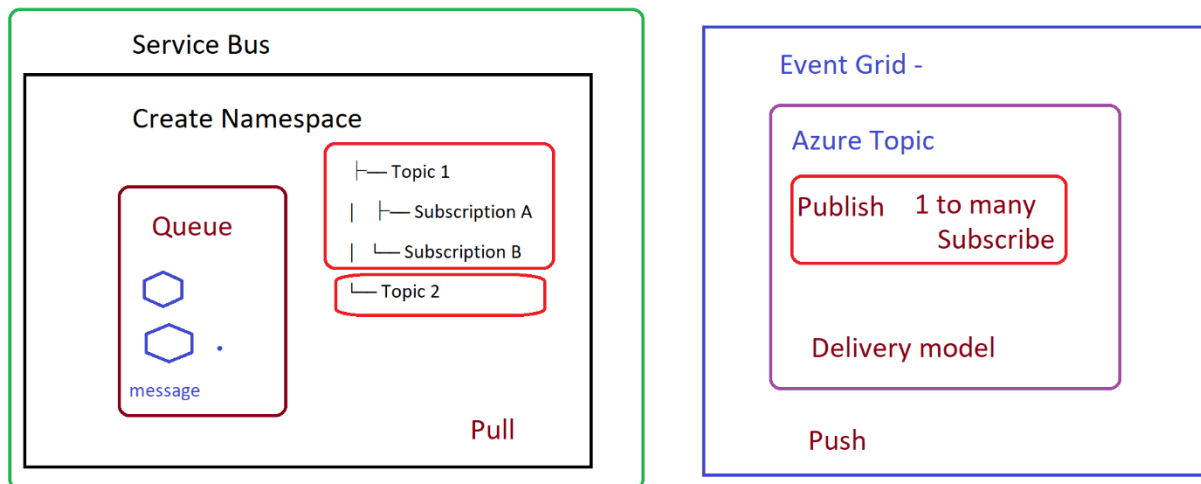
When to Use What?

Scenario	Use This
Need rich features like message deferral, dead-lettering, sessions, filtering	Azure Service Bus
Need simple decoupling for background tasks, no frills	Queue Storage
Need to ingest millions of telemetry messages per second	Event Hubs
Need to route events to multiple subscribers , with filters	Event Grid
Want to store queued messages in a durable, cheap way (e.g., for retries)	Queue Storage

Extra Notes

- **Service Bus** supports **Topics**, enabling **1-to-many pub-sub** like Event Grid, but with full message processing and filtering features.
- **Queue Storage** is best for **simple, cheap, durable messaging** where you manage retries and failures.
- **Event Grid** is ideal for **event-based orchestration**, like notifying subscribers when a blob is uploaded.
- **Event Hubs** is designed for **big data** — firehose-style ingestion for streaming analytics platforms like Azure Stream Analytics or Databricks.

Let me know if you want example use cases or diagrams to visualize how these might work together!



Service Bus Message Unit & Billing

◆ Is 1 Messaging Unit (MU) = 1 Message up to 1MB?

No, a **Messaging Unit (MU)** in **Azure Service Bus Premium** is **not** a per-message billing unit.

Instead, it's more like a **slice of dedicated capacity**, not a count of messages. You're **not charged per message**, but **per hour per MU**, regardless of how many messages you send — up to the throughput limits of that unit.

◆ What is a Messaging Unit?

A **Messaging Unit (MU)** is a measure of **dedicated compute, memory, and I/O** capacity allocated to your **Premium Service Bus namespace**.

- 1 MU can handle:
 - **~1,000–2,000 messages/second** (depending on size and workload)
 - Messages **up to 1MB**
 - **Unlimited** message operations (within MU limits)

You **scale up** by provisioning more MUs:

- 1 MU = baseline capacity
- 2 MUs = doubled capacity
- 4 MUs = quad capacity

💡 Think of it as **buying reserved server capacity** to process messaging workloads efficiently and with high isolation/performance.

◆ Do You Pay More for Larger Messages (e.g., 1MB)?

Yes, **message size still matters**, even in Premium:

Message Size	Billing Impact
≤ 256 KB	1 message operation
> 256 KB (up to 1MB)	Counted as multiple operations (for size only in Basic/Standard)
Premium Tier	No per-message billing — flat hourly MU fee

So:

- **Basic/Standard** tiers will **charge multiple operations** if message size > 256 KB.
 - **Premium** tier absorbs this — but you're **paying for MUs by the hour**, so large messages consume more of your allocated MU resources.
-

✓ Bottom Line

- **Messaging Units ≠ individual messages**
- You're **not billed per message** in Premium, but **per hour per unit**
- Premium gives you **predictable costs** and **dedicated performance**, not pay-as-you-go per operation like Basic/Standard

Would you like a calculator or sample scenario for choosing MUs based on expected traffic or size?

Yes — you're spot on. Let's break it down cleanly:

◆ Is a Namespace the Parent to Queues and Topics?

✓ Yes, in Azure Service Bus, a **Namespace** is the **top-level container**.

Think of it like this:

Service Bus Namespace

├— Queue 1

├— Queue 2

├— Topic 1

| └— Subscription A

| └─ Subscription B

└─ Topic 2

A **namespace** holds:

- **Queues**
- **Topics** (and their Subscriptions)
- Shared settings like:
 - Authorization rules (Access Control)
 - Metrics
 - Geo-replication (if Premium)

◆ How Many Namespaces or Queues Can You Have?

Here are the **limits** (as of current Azure docs):

Entity	Limit
Namespaces per Azure Subscription	~1000 (soft limit; can request more)
Queues per Namespace	10,000
Topics per Namespace	10,000
Subscriptions per Topic	2,000
Rules per Subscription	100

💡 Note: Limits may differ slightly between **tiers (Basic, Standard, Premium)** and some can be increased via Azure Support.

◆ Can You Have Multiple Namespaces?

✅ Yes, and you often **should**:

- For **isolation** (e.g., dev/test/prod)
 - To avoid **limit collisions**
 - To deploy in **different regions**
 - For **multi-tenancy** or performance scaling
-

Summary

Term	Role	Notes
Namespace	Container for all entities	Think of it like a messaging "server"
Queue	Holds messages (1:1)	FIFO, supports sessions, DLQs
Topic	Pub-sub messaging (1:many)	With filters and rules
Subscription	Listener to a topic	Can filter which messages it gets

Would you like a visual diagram or quick example of a real-world layout?

That's a sharp observation — and you're on the right conceptual track — but **Azure Event Grid Topics** and **Azure Service Bus Namespaces** are actually **entirely separate services** under the hood, even if they **seem similar** in structure.

Let's clarify the overlap and key differences:

◆ Are Event Grid Topics and Service Bus Namespaces the same?

✗ No. They are different services, with different goals and infrastructure.

Feature	Azure Event Grid	Azure Service Bus
Namespace concept?	✗ No namespace — just topics	✓ Has a namespace containing queues/topics
Messaging model	Push-based eventing	Pull-based messaging queue/topic model
Topic =	Custom event publisher (like a webhook hub)	Pub-sub message router inside a namespace
Delivery	Push to endpoints (HTTP, Function, Logic App)	Pull (or peek-lock) from queue/subscription
Latency	Near real-time (low latency)	Optimized for reliability & order guarantees
Durability	Best effort (24-hour retry, default 1–3 days)	Persistent with dead-lettering and sessions
Fan-out behavior	Native (1:N via subscriptions)	Via Topics + Subscriptions

◆ Why it Feels Similar

You're right that **Event Grid Topics** resemble **Service Bus Topics**:

- Both are **publish-subscribe** models.
- Both have **subscriptions** (Event Grid Subscriptions vs Service Bus Topic Subscriptions).
- Both **route messages/events** to multiple handlers.

But the core difference is **intent and infrastructure**:

Comparison	Event Grid Topic	Service Bus Topic
Designed for	Event-driven architectures	Messaging with strict control
Subscription model	Push to endpoint	Queue-like pull model
Retry behavior	At-least-once, timed expiry	At-least-once, persisted
Filtering	Event filters by type, subject SQL-like rules on subscriptions	

◆ So What's the Takeaway?

- **Event Grid Topic**: Lightweight, reactive, web-native event publisher.
- **Service Bus Namespace**: Heavy-duty, enterprise-grade messaging broker.

✓ Both support pub/sub, but:

- Event Grid is **event distribution**
- Service Bus is **message routing & persistence**

Let me know if you want a table comparing **Event Grid**, **Event Hub**, and **Service Bus** — that's where a lot of confusion stems from!

Great question — **Azure Event Hubs** is the third player in Azure's messaging/eventing trio, and it fills a **high-throughput, streaming** role that sits **between Event Grid and Service Bus** in terms of purpose.

Here's a clean breakdown of how **Event Hubs** fits into the mix:

▲ Azure Messaging/Events Services Summary

Feature	Event Grid	Event Hubs	Service Bus
Purpose	Event routing (notifications)	Real-time event streaming/logs	Reliable enterprise messaging

Feature	Event Grid	Event Hubs	Service Bus
Pattern	Push (1:N fan-out)	Ingest and stream (1:many consumers)	Pull/push queue/topic-based
Delivery	Push to endpoints (HTTP)	Pull by consumers (Kafka-like)	Pull/peek-lock by receivers
Latency	Low (milliseconds)	Low (ms to seconds)	Medium (configurable)
Retention	Short (1–3 days by default)	Configurable (up to 90 days)	Persistent until read
Ordering	No (unless 1 topic & single event handler)	Yes (per partition)	Yes (FIFO, sessions supported)
Scale	Millions of events/sec	Millions/sec per partition	Thousands/sec
Use Case	Lightweight pub-sub	Big data ingestion, telemetry, logs	Business logic integration, workflows
Consumer Model	Event handlers (push)	Stream processors (pull via offset)	Clients/apps consuming messages
Built-in Retry	Yes, 24–72h delivery attempts	Checkpointing & replays	DLQ, retries, transactions

◆ What Exactly Is Event Hubs?

✓ Event Hubs is:

- A **high-throughput event ingestion pipeline**
- Modeled like **Apache Kafka**
- Good for **real-time analytics**, telemetry, IoT, log ingestion

It supports:

- **Partitioned** message storage
- **Offset-based** read (like a durable stream)
- **Capture** to Azure Blob/Data Lake for archiving
- Integration with tools like:
 - Azure Stream Analytics
 - Apache Spark
 - Kafka-compatible clients
 - Azure Functions (via bindings)

◆ When to Use What?

Scenario	Best Choice
IoT telemetry ingestion	✓ Event Hubs
Audit logs / telemetry to storage	✓ Event Hubs
"Something happened" notification	✓ Event Grid
Email / payment processing queue	✓ Service Bus
Ordered, transactional workflows	✓ Service Bus
Distributed pub-sub w/ filtering	✓ Event Grid

 TL;DR Mental Model

Service	Like...	Analogy
Event Grid	Email notification system	"Notify me when X happens"
Event Hubs	Log file or Kafka pipeline	"Stream every change for later"
Service Bus	Enterprise mailroom	"Deliver messages reliably"

Let me know if you want a **diagram** showing how all three interconnect or an **example solution** combining all of them — that's a powerful pattern!

Azure Storage Queues vs. Service Bus Queues

Offer different features and are suitable for different scenarios. Storage queues are simpler, more cost-effective, and good for basic asynchronous message passing. Service Bus queues offer more advanced features like pub/sub, transactions, and complex routing, making them suitable for more sophisticated messaging scenarios.

[Compare Azure Storage queues and Service Bus queues - Azure Service Bus | Microsoft Learn](#)

Containers Including Docker

1. [Containers in Visual Studio Code](#)
2. [Get started with container apps in Visual Studio Code | Microsoft Learn](#)
3. A container is a process on your computer. It's isolated from all other processes on the host computer. That isolation uses kernel namespaces and control groups.
4. A container uses an isolated filesystem. This custom filesystem is provided by a container image. The image contains everything needed to run an application, such as all dependencies, configuration, scripts, and binaries
5. [Tutorial - Quick Docker Image Build with Azure ACR Tasks - Azure Container Registry | Microsoft Learn](#)

Azure Container Registry Tasks (ACR Tasks)

6. [Automate Container Builds with Azure Container Registry Tasks - Azure Container Registry | Microsoft Learn](#)
 - a. Enable automated builds triggered by source code updates, updates to a container's base image, or timers.
7. [Sample Azure Container Registry \(ACR\) Tasks - Azure Container Registry | Microsoft Learn](#)

Great question, Tim! You're absolutely right that both **Azure Container Instances (ACI)** and **Azure Container Apps (ACA)** pull container images from a registry (like Azure Container Registry or Docker Hub), similar to how you'd copy files to a web server before serving them. But the big difference lies in **what happens after the image is pulled**—the runtime environment and orchestration model.

Here's the breakdown:

✓ Azure Container Instances (ACI)

- **What it is:** ACI is a lightweight service for running containers directly on Azure without managing VMs or orchestrators.
- **How it runs containers:**
 - It spins up an isolated container (or a small group in a container group) on Azure's infrastructure.
 - There's **no orchestration layer**—each container group is standalone.
- **Use case:**
 - Great for short-lived tasks, batch jobs, or simple microservices.
- **Scaling:**
 - Manual or via external automation (e.g., Logic Apps, Functions). No built-in autoscaling.
- **Networking:**
 - Basic networking options, but not as advanced as ACA.

✓ Azure Container Apps (ACA)

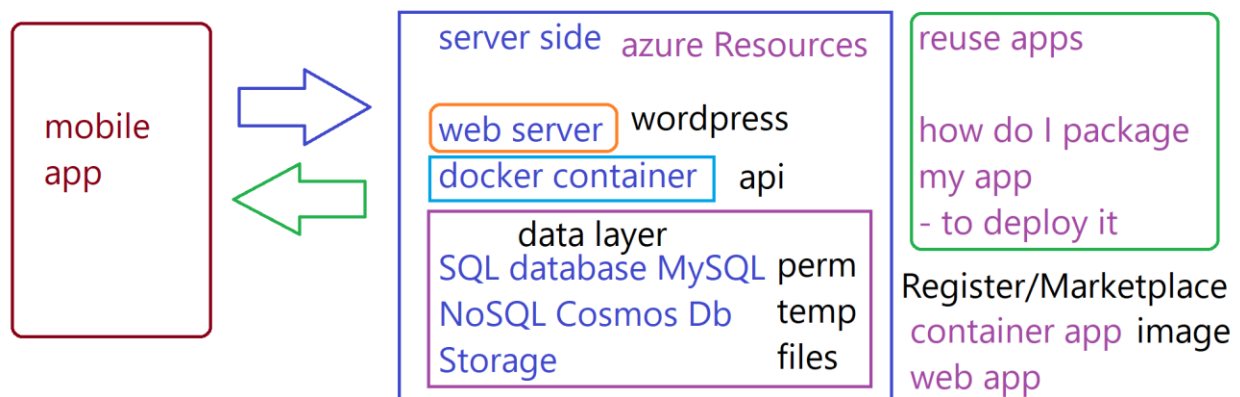
- **What it is:** ACA is a **managed container orchestration platform** built on Kubernetes and Envoy, but abstracted so you don't manage clusters.
- **How it runs containers:**
 - Containers run inside a **Kubernetes-based environment** with features like revisions, traffic splitting, and Dapr integration.
 - ACA adds **autoscaling (including KEDA)**, secrets management, and service discovery.
- **Use case:**
 - Ideal for microservices, event-driven apps, and apps needing scale-to-zero.
- **Networking:**
 - Advanced networking, ingress, and service-to-service communication.

🔑 Key Difference

Think of ACI as “just run this container now,” while ACA is “run this container as part of a scalable, orchestrated application.”

Both pull images from a registry, but ACA adds orchestration, scaling, and app lifecycle management.

Would you like me to **draw a quick diagram comparing ACI vs ACA architecture** (showing where the container image fits in and what extra layers ACA adds)? Or should I **write a concise analogy comparing them to web hosting models**?



Deployment Slots Swapping

[Swap deployment slots in Azure App Service](#) | [Develop solutions for Microsoft Azure](#)

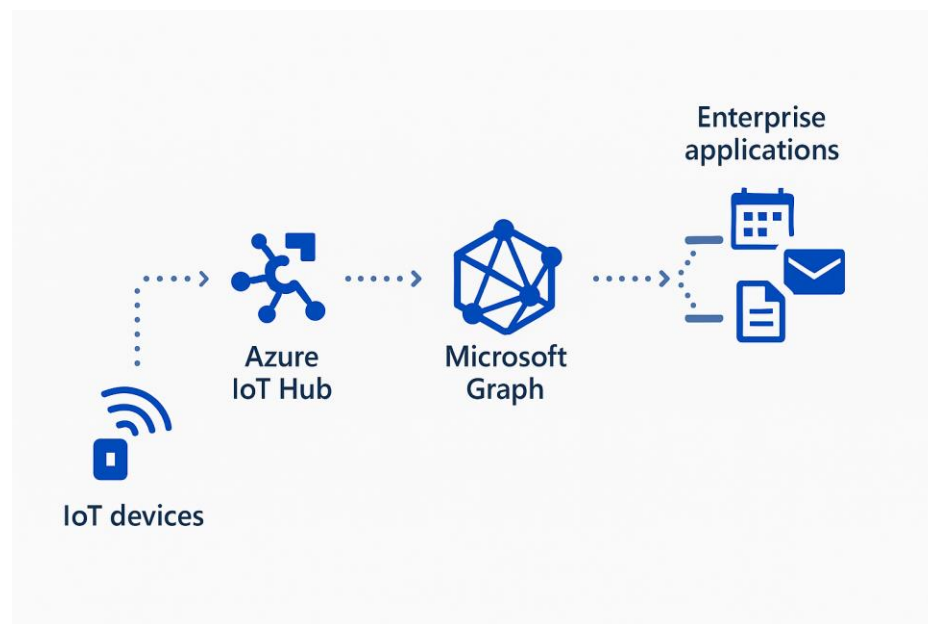
What is Microsoft Graph?

Microsoft Graph is a **RESTful API and SDK ecosystem** that provides a unified endpoint (<https://graph.microsoft.com>) to access data and services across Microsoft 365, Windows, and Enterprise Mobility + Security.

It acts as a **gateway to organizational data** like users, devices, files, calendars, and more.

Think of it as:

- **One API to rule them all** for Microsoft cloud services.
- It uses **OAuth 2.0 tokens** for secure access.
- Works with **delegated** (user) and **application** (daemon/service) permissions.



Graph's Place in IoT

IoT (Internet of Things) involves **connected devices** generating data and interacting with cloud services. Microsoft Graph fits into IoT scenarios as the **bridge between device identity, user context, and enterprise data**.

Key Roles of Graph in IoT

1. Device Identity & Management

- IoT devices often register in **Microsoft Entra ID** (formerly Azure AD).
- Graph API can query device objects, compliance status, and conditional access policies.
- Example: GET /devices or GET /me/ownedDevices.

2. Integration with Digital Twins & Azure IoT

- IoT solutions use **Azure IoT Hub** for telemetry.
- Graph complements this by linking **device identity** to **user identity** for enterprise workflows.
- Example: A smart badge device → Graph → user calendar → meeting room automation.

3. Security & Compliance

- IoT endpoints need secure access to organizational resources.
- Graph provides **conditional access**, **Intune device compliance**, and **role-based access**.
- Example: Only compliant IoT sensors can upload data to SharePoint via Graph.

4. Data Enrichment

- IoT telemetry can trigger workflows that use Graph data:
 - Fetch user profiles (/users)
 - Update Teams status (/me/presence)
 - Store logs in OneDrive or SharePoint (/drive/items)

5. Automation & Orchestration

- Combine IoT events with Graph to automate enterprise processes.
- Example: A factory sensor detects downtime → Graph sends Teams alert → updates Planner task.

Why AZ-204 cares?

- You'll need to know **how to secure IoT apps** using **Microsoft identity platform**.
- Understand **Graph endpoints** for device and user data.
- Implement **token acquisition** for IoT services (often app-only flow or managed identity).
- Use **Graph SDK** in Azure Functions or IoT Edge modules for automation.

Authentication MSAL.NET

AZ-204 expects you to:

1. **Authenticate & authorize** using Microsoft Entra ID (formerly AAD).
2. Use **MSAL.NET** to obtain tokens for different app types/flows.
3. Call **Microsoft Graph** securely using the **Graph SDK**.
4. Apply **least privilege**, consent, and **managed identities** where possible.

You'll use **MSAL.NET** to get an access token, then pass that token (or a TokenCredential) to the **Graph SDK** to call endpoints (e.g., users, mail, files).

Core Concepts You'll Need

App Types & Flows

- **Delegated** (user context): Desktop/Mobile/SPA/Web apps; scopes like User.Read.
 - Flows: **Auth Code** (+ PKCE), **Interactive**, **On-Behalf-Of**.
- **Application** (app context): Daemon/API/background jobs; app roles (.default).
 - Flow: **Client Credentials** (Confidential Client).

Permissions

- **Delegated scopes** (e.g., Mail.Read) require user sign-in, sometimes admin consent.
- **Application roles** (e.g., Mail.Read app permission via .default) require admin consent and no user.

Endpoints & Tenancy

- v2 endpoint: <https://login.microsoftonline.com/{tenantId}/oauth2/v2.0>
- Resource/audience for Graph: <https://graph.microsoft.com>
- Multi-tenant vs single-tenant matters for consent and authority configuration.

Reference

1. [Overview | Dapr Docs](#)
 - a. Incrementally migrate your existing applications to a microservices architecture, thereby adopting cloud native patterns such as scale out/in, resiliency, and independent deployments.
2. [Sign in with Azure CLI — Login and Authentication | Microsoft Learn](#)
3. [Install the Azure CLI for Windows | Microsoft Learn](#)

4. Routing traffic & View Cookies in Client Browser
5. [View, edit, and delete cookies - Microsoft Edge Development | Microsoft Learn](#)